Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol

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Submitted by:

Alcazar, Salmuel Whytte P.
Carillo, John Carl M.
Gonzales, Giorj Allen Q.
Quinal, Joven Gerald A.
Sison, Jahaziel B.

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Chapter 1
BACKGROUND OF THE STUDY

Renting a house is increasing rapidly in many countries. Even in developed nations such as Switzerland, Germany, and Denmark, a significant portion of citizens prefer renting over homeownership [1]. In developing countries, including India and Indonesia, low homeownership attainability makes renting a more viable option for many households [2].

In the Philippines, urban areas such as Manila, Cebu, and Davao have seen a similar trend. In a Consumer Finance Survey conducted by the Bangko Sentral ng Pilipinas (BSP), many Filipino families preferred renting accommodations rather than owning homes [3]. Several factors contribute to this trend. High property prices in urban areas make homeownership unaffordable for the average Filipino. Additionally, wages often grow slower than real estate prices, making renting a more practical and economical choice.

This study focuses on San Pedro, Laguna, known as the “gateway” or “dormitory town” of Metro Manila due to its proximity to the capital. As one of the most urbanized and densely populated cities in Laguna, San Pedro has numerous rental houses, apartments, and boarding spaces. Traditional methods of searching for rental housing in San Pedro, Laguna, remain inefficient, unreliable, and time-consuming. Therefore, this study aims to develop a web-based platform to improve the rental experience for residents.

Despite the increasing demand for rental housing, tenants face several challenges. Traditional house-hunting methods require visiting multiple properties to compare pricing and conditions, which is inconvenient and time-consuming. Social media platforms, such as Facebook, are sometimes used to facilitate rentals; however, these platforms remain largely unregulated.

Listings may be inaccurate, incomplete, outdated, or even fraudulent. Scammers often post realistic advertisements to deceive tenants into paying deposits for non-existent properties.

Furthermore, tenants often lack awareness of environmental and safety risks, such as flood-prone areas or crime hotspots, due to the absence of centralized risk visualization tools.

Currently, no reliable, secure, and verified platform exists specifically for San Pedro, Laguna residents. Existing solutions are often unmoderated, outdated, and insecure, exposing users to scams. Additionally, current platforms do not integrate safety-risk mapping or other local contextual information. This gap highlights the need for a secure, regulated web-based apartment management platform to improve security, accessibility, safety, and reliability in renting.

To address these issues, this project proposes a web-based apartment management system featuring **real-time communication, interactive mapping, and safe-zone visualization**. The platform will provide an efficient, secure, and reliable solution for both tenants and landlords.

Primary stakeholders include tenants seeking accurate, secure, and updated rental information, system administrators responsible for maintaining and updating the platform, and landlords who wish to list and manage their properties through a legitimate, verified platform.

This research aligns with the United Nations Sustainable Development Goals (SDGs). It supports SDG 11 (Sustainable Cities and Communities) by promoting accessible and reliable housing search, SDG 9 (Industry, Innovation, and Infrastructure) through the development of innovative technology for apartment management, and SDG 16 (Peace, Justice, and Strong Institutions) by fostering transparency and reducing rental-related scams.

The purpose of this project is to develop a comprehensive web-based apartment management platform integrating tenant and landlord functionalities. The system aims to enhance the tenant experience by providing a convenient, reliable, and efficient house-search process. It will offer accurate and updated property listings, real-time communication between tenants and landlords, location mapping for easy navigation, and a secure online payment gateway. Overall, the platform seeks to improve trust, transparency, and efficiency in the rental housing process.

Objectives of the Study

The objective of this project is to develop Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol. This project solves the limitations of traditional house hunting, online scams, and safety levels of the area. It will include features such as real-time communication, area mapping for the availability of apartments in the preferred location and its safety level, and rental management. Specifically, aims to achieve the following:

1. To design an online platform with the following major features:

1.1 Tenant Modules

a. Searching options for apartments
b. Availability of up-to-date and correct unit information
c. Interactive locations mapping
d. Real-time Direct Landlord Communication System
e. Submission of rental applications
f. Barangay filtering feature helps tenants easily find apartments in their preferred locations.
g. Safe zone maps to determine the location's affordability and safety.
h. Maintenance Request Submission System.
i. Access to payment history records.

1.2 Admin Modules

a. Landlord Application Review System
b. Management of tenant and landlord accounts.
c. Searching and filtering of apartment listing
d. Send notification to users regarding disaster events around San Pedro.
e. Property Listing Moderation

1.3 System Modules

a. Secure login and role-based access
b. User-friendly interface
c. Data privacy and access control
d. To construct the platform as designed, integrating real-time communication tools and mapping to enhance user experience.

2. To test a system through integration, functionality, usability, and performance evaluation to ensure quality and efficiency.

3. To evaluate the system using appropriate software quality standards in terms of:

a. Functional suitability
b. Usability
c. Performance efficiency
d. Reliability
e. Security
f. Maintainability
g. Portability

Significance of the Study

The findings of this study will highlight the important benefits of a comprehensive apartment management platform for all key users. By examining the impact of a one-stop system with features for real-time communication and location mapping, this research aims to provide valuable insights that may lead to further innovations in property technology. The researchers believe this study will primarily benefit the following groups:

- **Landlords:** The system offers a tool for administrative tasks, which is expected to enhance efficiency in property management and tenant communication.

- **Tenants:** The platform provides a streamlined and user-friendly interface for managing rental responsibilities. It is designed to simplify the processes of property consultation, and communication with landlords.

- **The Real Estate Industry**: This project contributes to the digital transformation of the local real estate market by integrating technology into property management. The study aims to demonstrate the benefits of this approach, setting a precedent for industry practices, and encouraging further innovation in the PropTech sector.

- **The Researchers**: This study provides a practical application of conceptual understanding in system analysis, design, and software development. It offers a comprehensive hands-on learning experience, enabling researchers to develop and apply skills in database management and project management to a real-world problem.

- **Future Researchers**: The research will serve as a foundational study and a valuable resource for academics and developers. The project's architecture, challenges, and findings can inform and guide future research efforts in developing more advanced and comprehensive property management systems.

Scope and Delimitation of the Study

Scope

The platform will provide a centralized system for landlords to manage multiple properties, including features for tenant onboarding, rent collection tracking, and maintenance request management, while also offering tenants a dedicated portal to pay rent, submit maintenance requests, and communicate with their landlord. It will incorporate a real-time communication module to enable instant messaging between tenants and landlords for efficient and quick resolution of queries and issues, and it will integrate location mapping services to allow tenants and landlords in the City of San Pedro to visualize available properties, making search and management processes more efficient.

Delimitations

- The platform will be a website rather than a native mobile application, ensuring a seamless user experience across various devices through a single web-based interface.

- The system will not include advanced administrative features such as automated tenant background checks, credit scoring, or the generation of legally binding contracts. It is designed primarily as a property management tool rather than a legal or financial services platform.

- The system does not process online rent payments or financial transactions. Tenants and landlords coordinate payments manually, but the system allows manual recording and tracking of payment history for reference purposes.

- The system allows tenants to submit maintenance requests and landlords to respond internally. However, it does not include automated integration with third-party repair service providers.

- The system will not include detailed reporting or analytics, such as occupancy rates, maintenance history summaries, or safety incident logs.

- The system provides verified and informative apartment listings, including environmental information such as flood risk, proximity to fault lines, hospitals, and evacuation centers. It does not exclude apartments located in hazard-prone areas, and the system does not provide real-time hazard monitoring or emergency response services. Its purpose is to ensure tenants are informed about potential risks to support informed decision-making.

- This project will be developed as a school-based initiative and will not function as a public-facing website accessible to external audiences or individuals outside the school premises.

- This study focuses on the design and development of the Safe-Zone Apartment Management System, limited to apartment owners, tenants, and administrators within San Pedro, Laguna, to address local rental management, communication, and safety concerns.

- The system's notification feature includes internal system and email notifications only and does not support SMS or other external communication methods.

Chapter 2
REVIEW OF RELATED LITERATURES AND SYSTEMS

This chapter shows the relevant review of related literature and studies needed for the design and development of the system.

Technical Background

This project entitled “Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol” features real-time communication and geographical information systems with tenant and landlord modules to make renting and managing apartments more convenient, reliable, secure, and safe. It includes management functions with features that are created to help tenants and landlords to perform rental activities more easily.

Here's how it will work and the features it may include:

- **Tenant Modules:** This platform will provide tenants with filtering options for apartments, updated apartment information, location mapping, real-time communication with landlords, and the ability to submit rental applications online.

- **Landlord Modules**: Landlords can post and update property listings, manage tenant detail, and communicate with tenants in real time and send rental reminders.

- **System Modules**: The system ensures secure login, role-based access, data privacy, geographical information system, time proximity, and an interactive dashboard that consolidates key information for both tenants and landlords.

Technologies, Programming Languages, Tools, and API Used

1.Programming Languages

- **HTML and CSS** - used to build and design the website to make it more interactive and user-friendly

- **Javascript** - used to enhance the interactivity of the platform. JavaScript will be used to enhance the platform's interactivity. It allows developers to implement dynamic features such as real-time notifications, synthesis of the related literature and studies/system in research interactive maps, filtering options, and instant messaging between tenants and landlords, improving the overall user experience.

- **PHP** - is a popular server-side scripting language that will be used to develop the backend of the platform. It will handle server-side operations such as processing user requests, managing sessions, and connecting to the database. PHP enables dynamic content generation and allows the system to efficiently manage interactions between tenants and landlords in real time.

- **MySQL** - an open-source relational database management system that stores and organizes essential information such as property listings, user accounts, tenant records, and transaction histories. Using MySQL ensures that data is structured, secure, and easily retrievable, providing a reliable foundation for the platform's operations.

2. Application Programming Interface (API)

- **OAUTH (Open Authorization)** - an open-standard framework for authorized access delegation, allowing websites or applications to access user information from other services without requiring passwords.

- **Google Maps -** is a set of application programming interfaces (APIs) and software development kits (SDKs) that allow developers to embed Google Maps, locations,

and interactive geographic data into web pages and mobile applications. It enables features like custom markers, 3D building views, route planning, and searching for over 200 million places.

- **PHPMailer** — a full-featured, open-source email sending library for PHP that allows developers to send emails directly from a web application. It supports SMTP authentication, HTML email content, file attachments, and multiple recipients, making it a reliable alternative to PHP's built-in mail() function.

3. Software Tools

- **Visual Studio Code -** main development environment for building the frontend and backend
- **Xampp -** provides the local server environment for PHP and MySQL
- **phpMyAdmin** – manages the MySQL database easily
- **Google Chrome** – Used for testing and debugging the web interface
- **Figma -** a cloud-based design and prototyping tool for creating wireframes and interactive prototypes of the platform. It enables developers and designers to collaborate efficiently and ensure the user interface is visually appealing, intuitive, and user-friendly.

These tools and technologies collectively provide a solid, scalable, and secure foundation for the web-based safe-zone apartment management platform. by leveraging modern web development frameworks, real-time communication tools, and interactive mapping technologies, the system will deliver a convenient, reliable, and efficient solution for tenants and landlords.

Foreign and Local Related Literatures

The Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol is rooted in research that emphasizes the significance of technology in managing properties. Studies on tenant safety, verified property data, and online fraud prevention directly inform the system's core features and safety protocols. These studies provide a foundation for designing a reliable, tenant-focused solution that enhances safety and prevents fraud.

Foreign Related Literatures

Real-time communication enables tenants and landlords to interact efficiently without relying on physical meetings or delayed responses. A study on Smart Management Services for high-rise apartment residents in Beijing, China found that real-time communication tools, instant notifications, and digital service portals significantly improve tenant satisfaction [4]. These features help streamline communication, reduce response time, and improve service delivery in residential management systems. These findings support the design of Safe-Zone, which integrates similar communication and notification features to enhance tenant engagement, safety, and convenience.

Online rental listing platforms have become a major source of real-time housing market data, with studies on platforms such as Zillow, Trulia, and Craigslist showing variations in rental listings, pricing, and information accuracy. However, these platforms often face issues related to inconsistent data, lack of standardization, and limited transparency in property information [5]. Such limitations may lead to misinformation and reduce tenant trust when searching for rental properties online. This supports Safe-Zone’s approach of providing a unified and verified property database to ensure accurate and reliable information for tenants and landlords.

In addition, GIS-based flood hazard assessment studies demonstrate the

effectiveness of spatial analysis and proximity-based risk modeling in identifying high-risk areas using environmental factors such as elevation, slope, drainage density, and land use. Although these studies focus on environmental hazards, they show how geographic data can be used for predictive risk assessment and decision-making [6]. The integration of multi-criteria spatial analysis allows for better identification of vulnerable locations and supports proactive planning. Similarly, Safe-Zone applies GIS-based proximity mapping to assess tenant safety and identify potential risks in apartment environments.

Further studies on data-driven housing safety systems show how centralized datasets such as parcel records, rental registries, housing violations, and eviction data can be used to identify high-risk properties and improve regulatory monitoring [7]. These systems demonstrate the value of integrating multiple data sources to create a more comprehensive view of housing safety conditions. By linking administrative and spatial data, authorities can make more informed decisions and implement targeted interventions. This highlights the value of combining such approaches, which Safe-Zone adopts to improve apartment safety monitoring and decision-making.

Lastly, the rapid expansion of digital platforms has increased exposure to online risks such as fraud, misrepresentation, and deceptive listings, particularly in online housing transactions. Studies emphasize that these risks can significantly affect consumer trust and decision-making in digital marketplaces [8]. As a result, verification mechanisms and secure system designs are essential to reduce user vulnerability and improve platform reliability. In response, Safe-Zone incorporates verified listings, secure communication channels, and real-time monitoring to enhance trust and ensure safer apartment hunting experiences.

Local Related Literature

In Region XII, Philippines, a cloud-based property management system was developed to improve the efficiency of real-estate operations. The system emphasizes secure tenant–landlord interactions through dedicated portals, real-time billing notifications, and access-controlled records. The study concluded that cloud-based platforms enhance trust between landlords and tenants by reducing errors associated with manual rent collection and record-keeping. These findings align with Safe-Zone by highlighting the importance of secure communication, accurate data management, and improved transparency in apartment rental systems [9].

Similarly, a local study on an Apartment Rental Management System for Celso Apartment in Cebu addressed common issues such as disorganized tenant records, inaccurate financial tracking, and inefficient communication between landlords and tenants. The system implemented real-time transaction monitoring and automated reminders to improve rent collection and administrative efficiency.

Results showed that digital platforms significantly reduce delays and improve overall property management performance. This further supports Safe-Zone in adopting automated processes and real-time updates to enhance operational efficiency in apartment management [10].

A study on a unified web-based Property Management System (PMS) highlights the increasing demand for efficient digital solutions in the real estate sector following the COVID-19 pandemic recovery. The system integrates property search, online payments, maintenance tracking, and centralized data management, resulting in improved communication and decision-making among stakeholders [11]. However, it primarily focuses on administrative efficiency and lacks real-time safety monitoring and location-based features. This limitation supports Safe-Zone's enhancement of traditional PMS functions by incorporating real-time proximity mapping and tenant safety protocols for improved security and responsiveness [11].

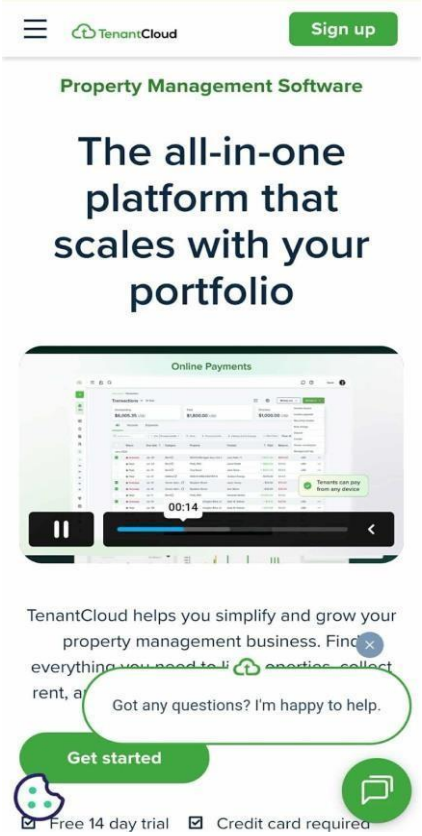
At the local governance level, a study on a Barangay Geographic Information System (B-GIS) shows how integrating GIS with management information systems improves documentation, spatial planning, and service delivery. The system addressed limitations of paper-based processes and achieved high ratings in usability, reliability, and performance based on ISO 25010 standards [12]. These findings demonstrate the effectiveness of GIS-enabled platforms in improving decision-making and community management. This supports Safe-Zone's use of GIS-based mapping to enhance apartment monitoring, tenant safety, and location-based risk assessment at the residential level [12].

A study on online scam experiences in Lubao, Pampanga highlights how the rapid growth of digital platforms increases exposure to cybercrimes such as online fraud. Victims were found to be mostly young adults with limited income and lower educational attainment, making them more vulnerable to deceptive schemes [13].

The study also reported significant emotional, financial, and social impacts, despite moderate awareness of online scams. These findings emphasize the need for stronger digital literacy, verification systems, and consumer protection measures. This supports Safe-Zone's goal of reducing fraudulent risks in apartment hunting through verified listings, secure tenant management, and real-time safety protocols [14].

Related Systems

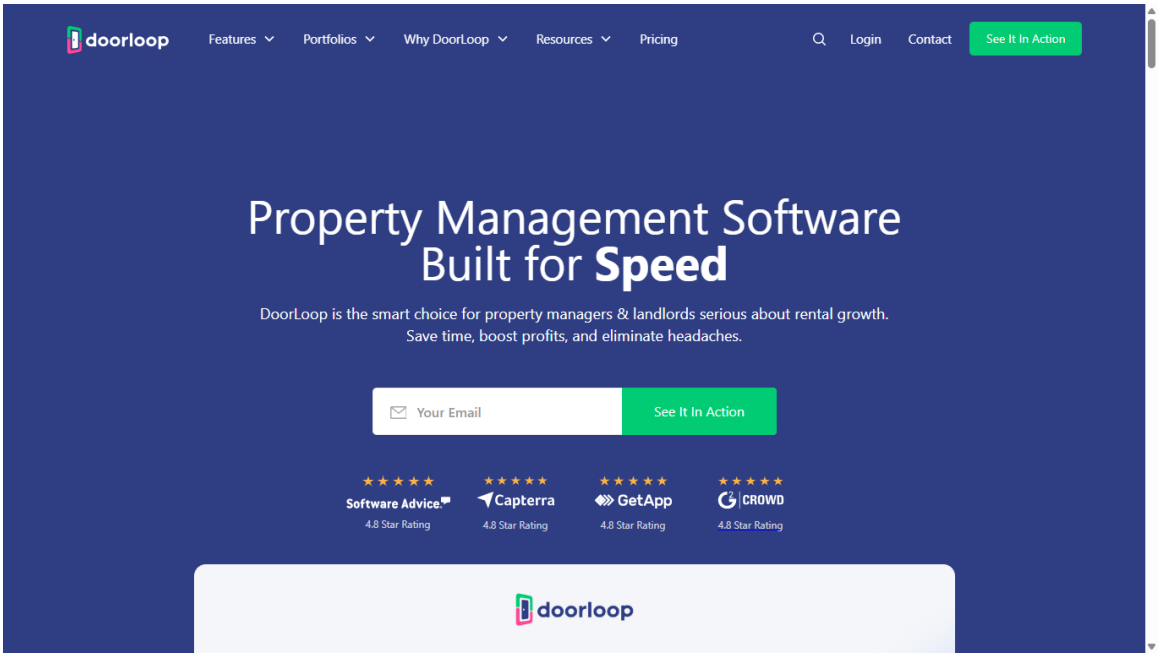
Foreign Systems



www.tenantcloud.com

Figure 1. TenantCloud

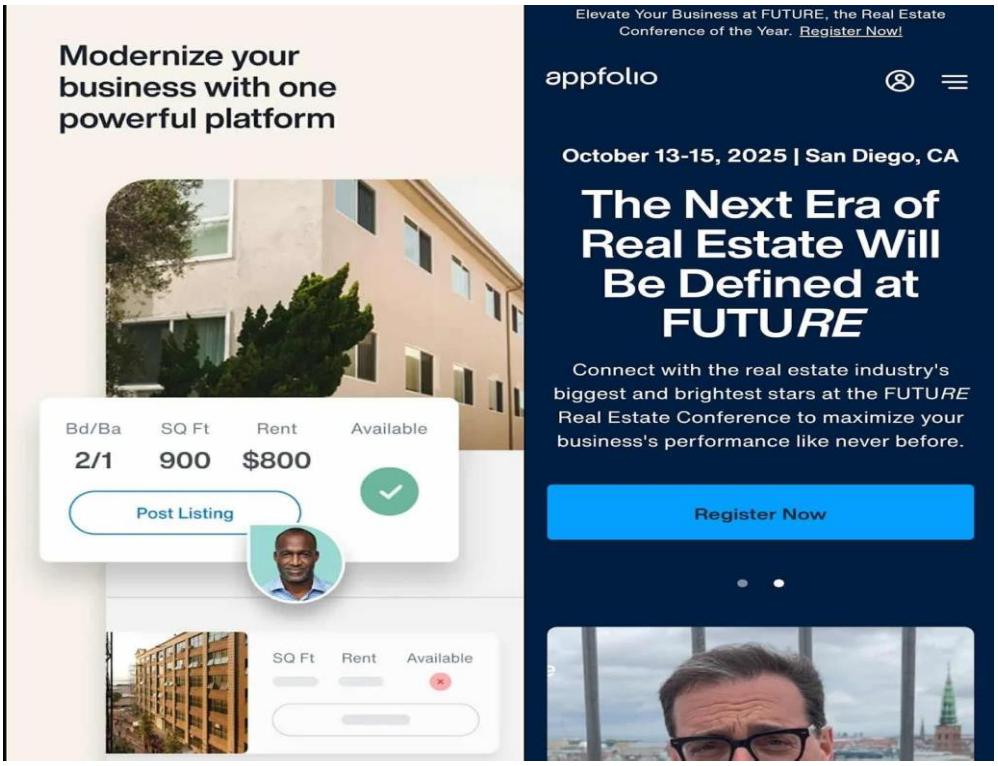
TenantCloud is a cloud-based platform that helps landlords manage properties and allows tenants to easily access rental services. Landlords can handle listings, rent collection, maintenance, and tenant records, while tenants can apply for units, pay rent, and communicate with landlords. Overall, it improves efficiency, transparency, and the rental experience for both parties.



www.doorloop.com

Figure 2. Doorloop

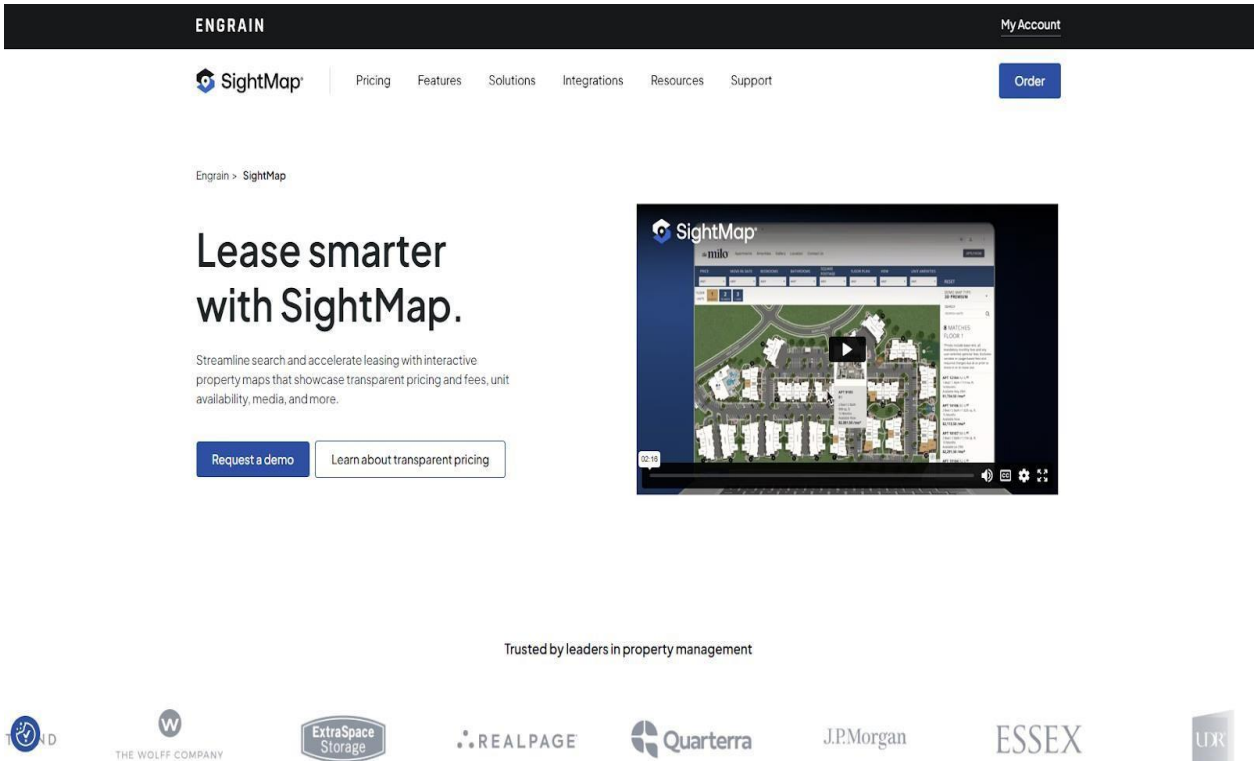
DoorLoop is a web-based property management system that integrates tenant and landlord features into a single platform. It enables landlords to automate rent collection, track maintenance requests, and screen tenants, while tenants can explore available apartments, pay rent online, and submit service requests. Real-time messaging tools ensure instant communication between tenants and landlords, minimizing delays and misunderstandings. By providing a streamlined workflow, DoorLoop reduces administrative tasks and enhances the overall rental experience.



www.appfolio.com

Figure 3. Appfolio

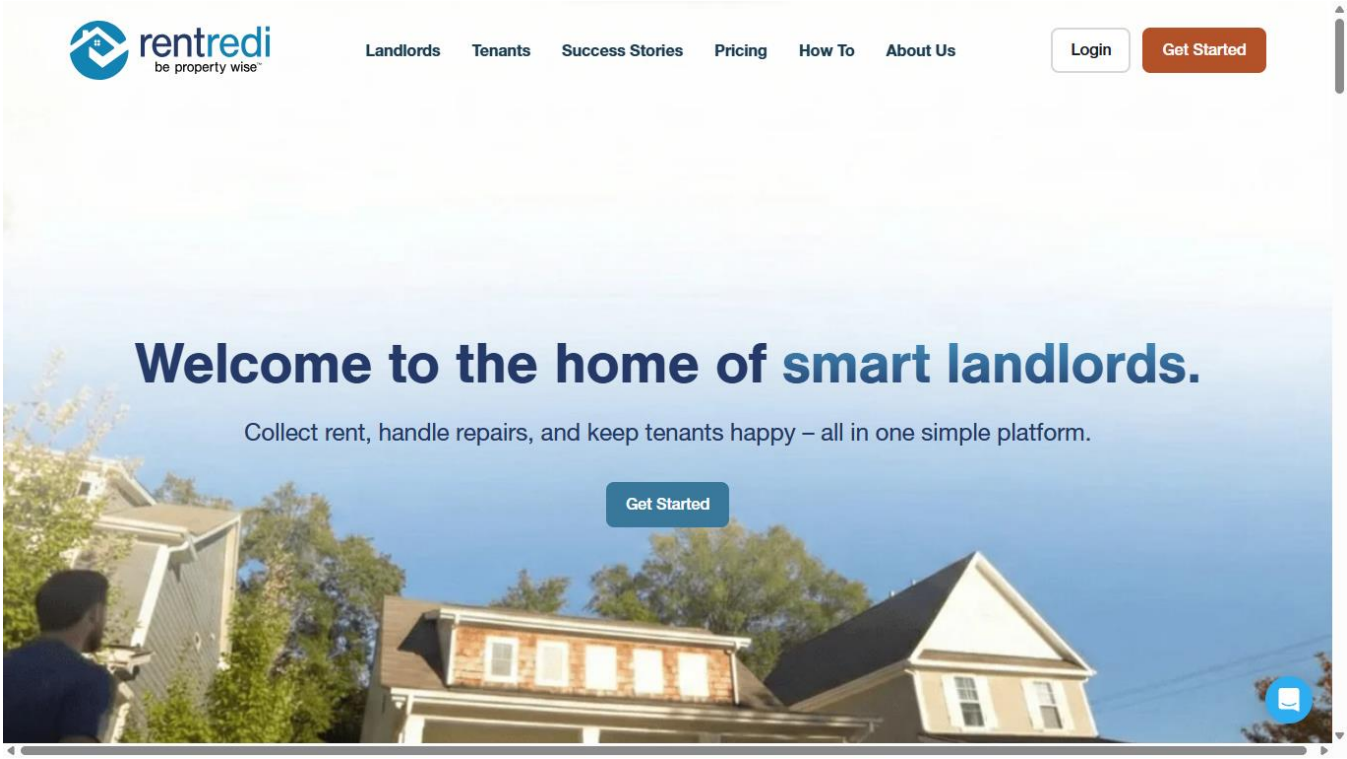
AppFolio is a cloud-based property management platform catering to residential and commercial properties. The system allows landlords to manage leases, collect rent, track maintenance issues, and store important documents digitally. Tenants benefit from a user-friendly interface for paying rent, submitting maintenance requests, and communicating with property managers. AppFolio's real-time updates and organized dashboards improve operational efficiency while providing tenants with transparency and convenience throughout the rental process.



www.engrain.com

Figure 4. Engrain SightMap

Engrain SightMap is an interactive web-based property mapping platform that provides tenants with visual representations of available rental units, pricing, and amenities. Landlords can update listings in real time, while tenants can explore units using interactive maps and detailed property information. The platform enhances decision-making by allowing tenants to evaluate accessibility and convenience before applying for a unit. With its real-time updates and interactive maps, SightMap creates a more engaging and transparent rental experience for both landlords and tenants.



www.rentredi.com

Figure 5.Rentredi

RentRedi is a comprehensive property management platform designed to streamline the rental process for both landlords and tenants. It offers features such as online rent collection, tenant screening, maintenance request tracking, and real-time messaging. Landlords can manage their properties efficiently, while tenants benefit from a user-friendly interface for submitting maintenance requests, making payments, and communicating directly with property managers. The platform's mobile-friendly design ensures accessibility and convenience for all users. By centralizing these functions, RentRedi enhances operational efficiency and improves the rental experience for both parties.

Local Related Systems

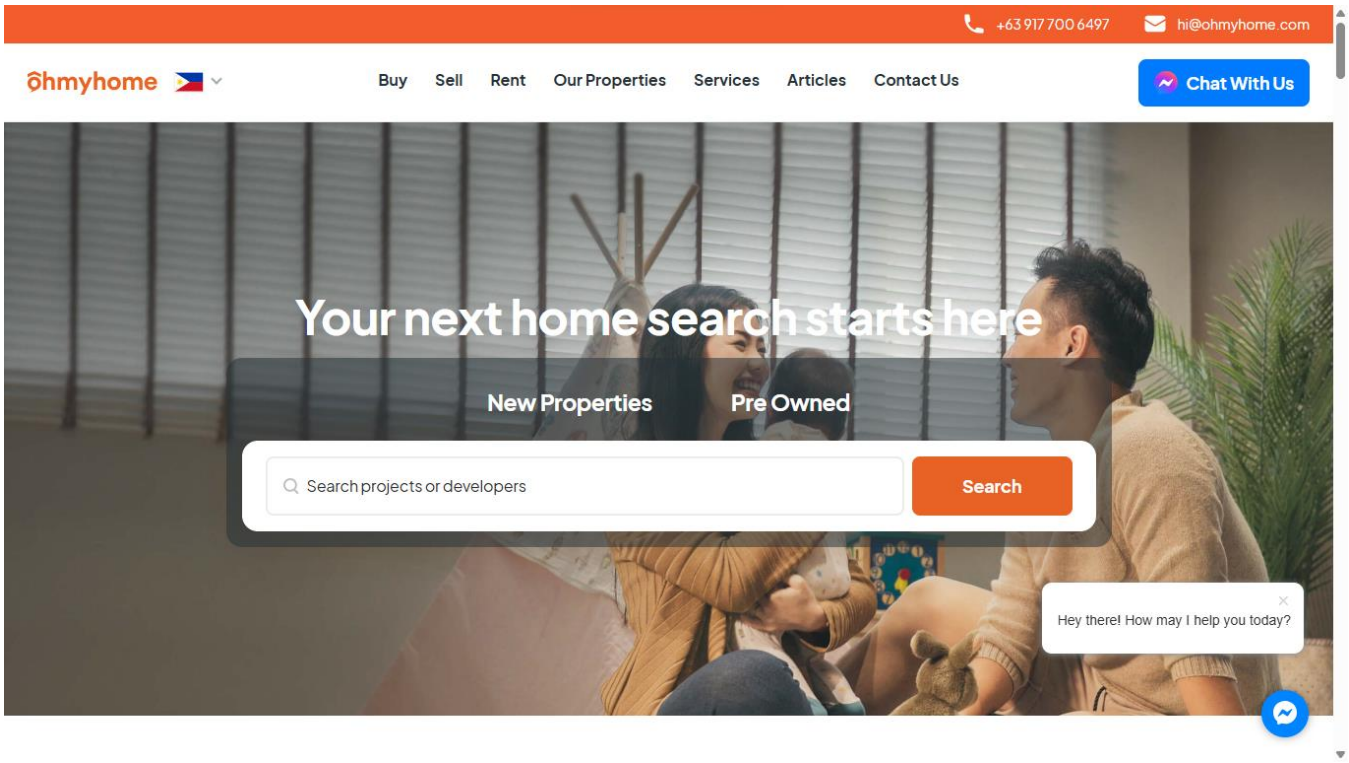


Figure 6. Ohmyhome

https://ohmyhome.com/en-ph/

Founded in 2016 by sisters Rhonda Wong (CEO) and Race Wong (CPO), **Ohmyhome** was started with one goal – to make housing transactions simple, efficient and affordable. This comes with a firm belief that everyone, regardless of socioeconomic status, should have access to quality and reliable home transacting resources.

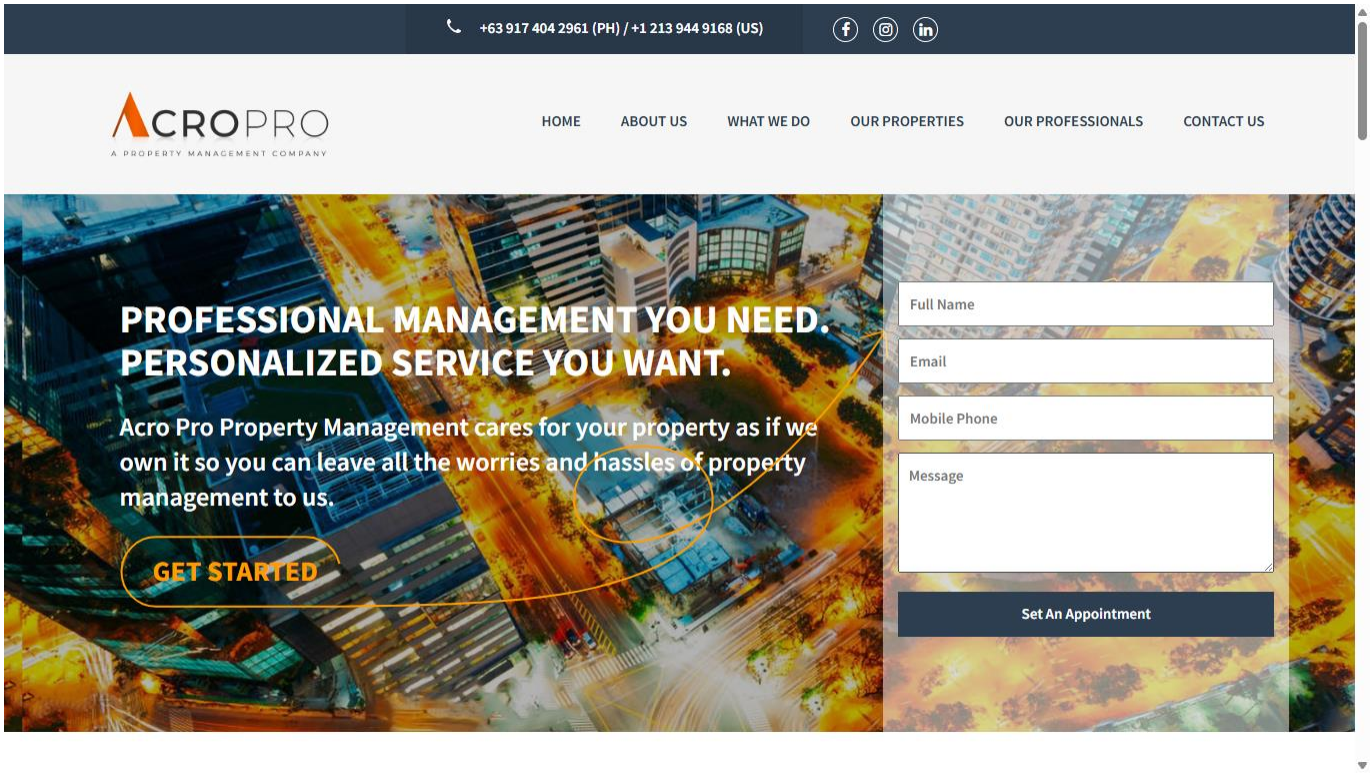


Figure 7. AcroPro Landing Page

https://acropro.ph/

Acro Pro Philippines (Acropro PH) is a Manila-based, LinkedIn Philippines specialized property management company established in 2019 to handle residential and commercial assets, particularly for overseas owners. They offer comprehensive services, including tenant management, maintenance, renovations, and financial reporting.

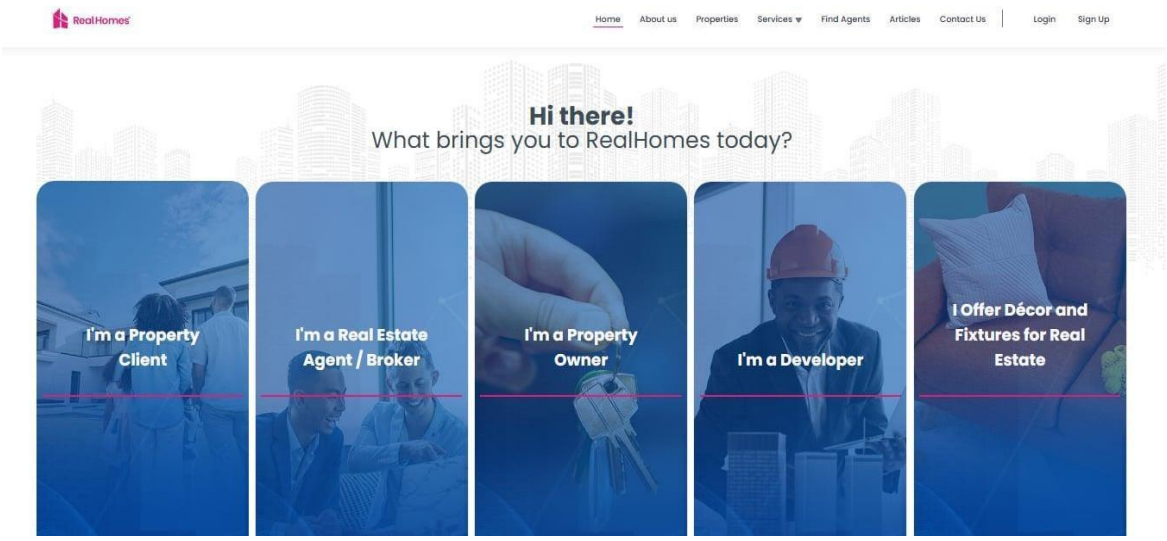


Figure 8. RealHomes

https://www.realhomes.ph/

RealHomes.ph is an all-in-one, end-to-end digital real estate marketplace based in the Philippines. It is designed to connect property buyers, renters, owners, agents, and developers, aiming to make real estate transactions faster, more secure, and easier.

Figure 9. Filipino Homes

https://filipinohomes.com/about.php

Filipino Homes is the Philippines' largest comprehensive real estate solutions portal and network, offering a one-stop-shop for buying, selling, and leasing properties. Based in Cebu with over 10,000–15,000 agents, it provides services like property management, forensics, insurance, and brokerage, aiming to support local and OFW clients.

Figure 10. IBoardLiving

https://www.iboardliving.com/about-us

iBoard Living is a Philippines-based, all-in-one "business-in-a-box" digital platform designed to modernize the rental property management industry. It connects tenants, property owners, and service providers to streamline finding, renting, and managing, specifically targeting boarding houses, apartments, and student accommodations with safe, verified, and tech-enabled solutions.

Synthesis of the Related Literature and Studies and Systems

The reviewed literature shows that digital platforms significantly improve apartment and property management by addressing issues such as manual record keeping, delayed payments, and ineffective landlord–tenant communication. Local studies in the Philippines have demonstrated that cloud- and web-based systems increase efficiency, accuracy, and trust through centralized records, automated notifications, and secure user access. Evaluations based on system quality standards further confirm that usability, reliability, and performance efficiency are essential factors for ensuring user satisfaction and effective system adoption.

However, several studies have revealed limitations in existing property management practices. Research on Philippine real estate development, particularly those targeting Overseas Filipino Workers, highlights the continued reliance on human-centered and labor-intensive approaches that lack digital tools for real-time monitoring, tenant engagement, and safety assurance. This gap indicates the need for technology-driven solutions to enhance transparency, accountability, and security. Additionally, local studies on Geographic Information Systems demonstrate that integrating spatial data improves planning, coordination, and service delivery, supporting the use of real-time mapping and proximity-based features in apartment management systems.

More broadly, the literature on digital platforms in the Philippines emphasizes their growing role across various sectors while underscoring the importance of proper system design, data governance, and user protection. Existing local platforms tend to focus on basic administrative or listing functions, whereas foreign systems are often not tailored to local community contexts. These findings highlight the need for localized and integrated solutions. In response, the proposed Safe-Zone system aims to combine apartment management, real-time proximity mapping, and tenant safety protocols into a secure and community-focused platform tailored to the needs of San Pedro, Laguna.

DEFINITION OF TERMS

Technical Terms

Apartment Management System

A software application designed to help landlords and tenants manage rental properties efficiently. It organizes property listings, tenant records, and landlord activities in one platform to reduce manual work and errors.

Application Programming Interface (API)

A set of tools and protocols that allow the system to communicate with other software applications. APIs make it possible to integrate services such as login verification, payment gateways, or map services.

Cloud-Based System

A system where data and services are stored on remote servers and accessed over the internet. This allows users to use the system anytime and anywhere without installing it on their devices.

Geographic Information System (GIS)

A technology used to collect, store, analyze, and visualize location-based data. In this study, GIS helps display apartment locations, nearby landmarks, and safety zones, providing tenants with better information for decision-making.

ISO 25010

An international standard used to measure the quality of software. It evaluates important features such as usability, reliability, performance, security, and functionality.

Real-Time Communication

A feature that allows users to send and receive messages instantly. This ensures that tenants and landlords can communicate quickly without waiting for delays or physical meetings.

Real-Time Proximity Mapping

A system function that shows the distance between apartments and nearby areas, such as schools, markets, or high-risk zones. It helps users assess convenience and safety when choosing a rental property.

Role-Based Access Control

A security feature that gives different permissions to users depending on their role, such as tenant, landlord, or system administrator. This ensures that users can only access information relevant to their responsibilities.

Safe-Zone Visualization

A mapping tool that highlights areas considered safe or unsafe based on crime reports, environmental hazards, or other risk factors. It helps tenants make informed decisions about where to live.

Web-Based Platform

An online application that can be accessed through a browser. It allows tenants and landlords to perform property management tasks without downloading or installing a separate application.

Operational Terms

Administrator

A system user responsible for managing accounts, verifying landlord and tenant information, and ensuring that the platform operates smoothly and securely.

Barangay

The smallest administrative division in the Philippines. The platform uses barangay data to help tenants filter apartment searches based on specific neighborhoods or locations.

Digital Platform

An online system that allows users to access services, share information, and perform tasks digitally. The Safe-Zone system is a digital platform designed to make apartment management and safety monitoring more efficient.

House-Hunting

The process of searching for available rental properties. Through the platform, tenants can filter apartments by location, price, safety, and other criteria, making the search process faster and more organized.

Landlord

A property owner or authorized representative who lists and manages rental properties on the platform. Landlords can update property information, track tenant details, and communicate in real time.

Local Government Unit (LGU)

The administrative body responsible for governing a city, municipality, or barangay. In this study, LGUs provide contextual information for safety, zoning, and community services.

Online Scam

Fraudulent activity conducted online to deceive tenants or landlords. Examples include fake listings, false property information, or requests for deposits for non-existent properties.

Property Management

The overall process of managing rental properties, including tenant coordination, record-keeping, rent collection, and maintenance monitoring.

Rental Application

A digital form submitted by tenants to apply for an apartment. It allows landlords to review applicant information and process approvals efficiently.

Rental Listing

A detailed description of an available apartment, including location, price, unit details, and images. The listing helps tenants choose the right property based on their preferences.

Tenant

A person who rents an apartment through the platform. Tenants use the system to search for apartments, communicate with landlords, and submit rental applications.

Tenant–Landlord Interaction

The communication and exchange of information between tenants and landlords. This includes inquiries, rental applications, payment coordination, and maintenance requests.

Verified Listing

A property listing that has been reviewed and approved by the system administrator to ensure accuracy, legitimacy, and safety for tenants.

CHAPTER 3

METHODOLOGY

Current Process and Situation

Currently, apartment renting in San Pedro, Laguna is still handled in a largely manual and scattered manner. Most available units are advertised through social media platforms, such as Facebook Marketplace, community group posts, or simple word-of-mouth referrals. Although these channels are easy to access, they often lack reliability. Listings may contain incomplete details, outdated availability, or misleading information, and in some cases, tenants encounter fraudulent posts. This situation makes it difficult for prospective renters to confidently assess whether an apartment is legitimate, safe, and suitable for their needs.

Once a tenant finds a potential listing, communication with the landlord usually occurs through personal messaging applications, text messages, or phone calls. Tenants must manually inquire about rent, rules, and availability, while landlords check availability using personal notes, memory, or informal records. Scheduling a unit visit is also done manually, requiring both parties to repeatedly exchange messages to agree on the date and time. This process is time-consuming and often leads to delays, missed messages, or scheduling conflicts.

After the physical visit, rental agreements, payments, and tenant records were handled offline. Transactions are commonly made in cash or through informal arrangements, and tenant information, such as contracts, payment history, and maintenance concerns, are recorded in notebooks, spreadsheets, or chat conversations. Without a centralized platform, important data can be easily misplaced, forgotten, or misunderstood. Maintenance requests may be delayed, payment tracking becomes difficult, and misunderstandings between landlords and tenants become more frequent.

This traditional setup highlights the need for a more organized and secure rental-management approach. Although the current process is familiar and widely used, it lacks efficiency, transparency, and tenant protection. The absence of a centralized and verified digital system allows misinformation, scams, and poor record management. As a result, tenants spend excessive time verifying listings and coordinating with landlords, while landlords face challenges in managing multiple properties and maintaining precise records. These issues emphasize the relevance of developing a digital apartment management system that can streamline rental processes, improve communication, and promote safety and accountability for both parties.

Problems and Challenges Identified

- **Unverified listings**

  Most apartment listings are posted on social media without any form of validation. Some posts are incomplete, outdated, or misleading, which exposes tenants to scams and wastes time verifying whether a unit is still available or even legitimate.

- **Manual record management**

  Landlords commonly store tenant information, payment records, and maintenance concerns using notebooks, chat messages, or basic spreadsheets. This manual approach increases the risk of lost data, inaccurate records, and difficulty in tracking multiple tenants or properties.

- **Inefficient communication**

  Communication between landlords and tenants relies heavily on informal platforms such as Facebook Messenger, SMS, or phone calls. Important details can be overlooked, messages can be missed, and there is no centralized communication channel to keep conversations organized.

- **Limited property visibility**

  Tenants have no efficient way to view or compare nearby apartments

based on location. Without integrated mapping or proximity-based tools, finding safe and accessible rental units becomes more difficult, especially for newcomers to the area.

- **Safety concerns and lack of area risk awareness**

  Tenants have limited access to reliable information regarding the safety of an apartment surrounding area. Existing rental listings do not indicate whether a property is near high-risk locations, poorly lit streets, or areas prone to incidents. Without mapped safe zones or proximity-based safety indicators, tenants are forced to rely on personal judgment or informal advice, which may be inaccurate or outdated.

- **Lack of transparency and verification**

  There are no established mechanisms to verify the identity and credibility of landlords or tenants. This lack of transparency reduces trust between both parties and increases the likelihood of fraudulent activities and disputes.

- **Time-consuming rental process**

  The overall rental process, from searching and inquiry to visitation and negotiation, requires significant time and effort. The absence of an integrated digital system results in repetitive steps, delays, and inefficiencies for both tenants and landlords.

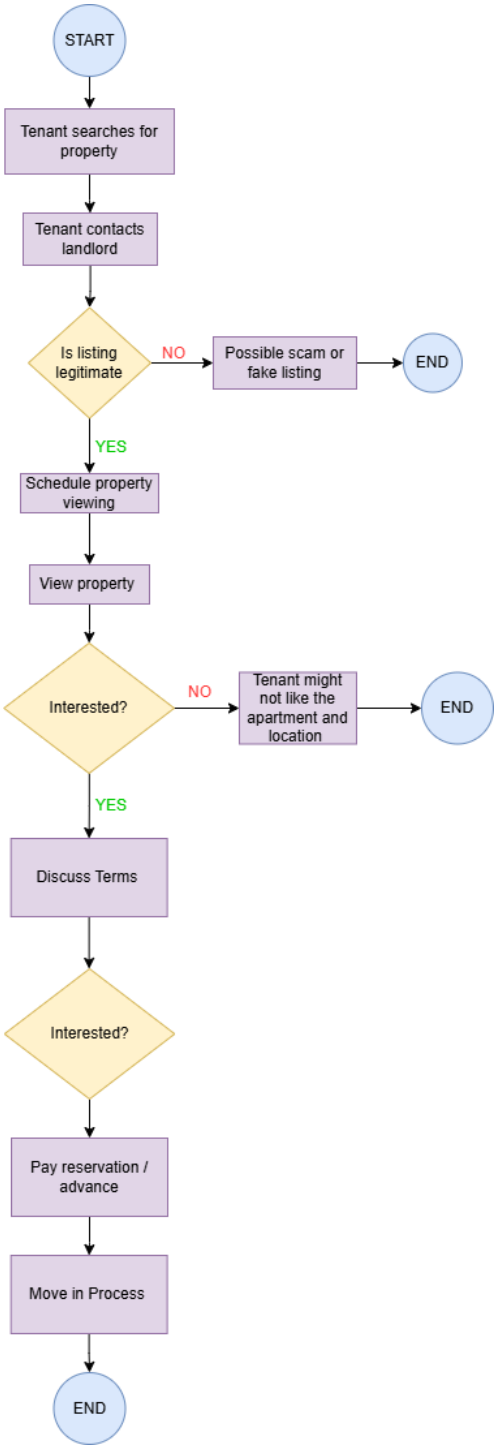


Figure 11. Flowchart for Common Apartment System

Current Proposed Process/System

The study proposes the design and development of **Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol** to improve the existing apartment rental process in San Pedro, Laguna. The proposed system aims to provide a centralized digital environment where landlords and tenants can manage rental-related activities more efficiently. By bringing listings, communication, and record management into a single platform, the system addresses the limitations of the current manual and fragmented setup.

The platform is designed to support verified apartment listings, allowing tenants to access accurate and up-to-date property information. Real-time communication features will enable direct and organized interaction between landlords and tenants, reducing misunderstandings and delays. In addition, integrated location mapping and safe-zone indicators will help tenants assess not only the availability of units but also the safety and accessibility of their surrounding areas.

Overall, the proposed system seeks to replace informal and unverified rental practices commonly found on social media with a reliable, user-friendly, and data-driven solution. By improving transparency, security, and efficiency, the platform aims to benefit both landlords and tenants while promoting safer and more informed rental decisions.

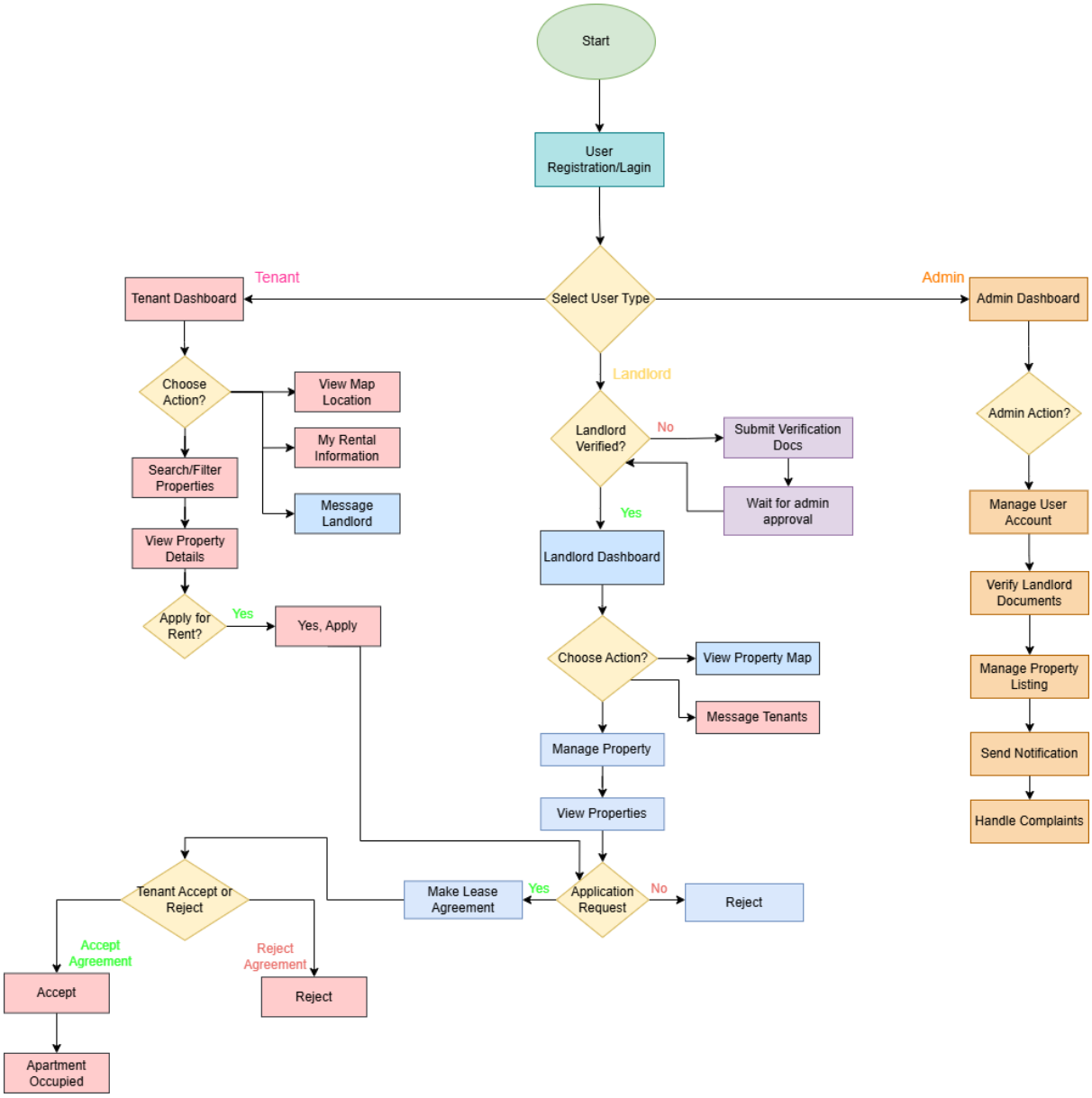


Figure 12. Flowchart of Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol

For the Administrator

The Administrator Dashboard serves as the control center of the platform. This allows the administrator to monitor user activity, manage property listings, and verify account authenticity. The administrator can view reports and oversee all interactions between tenants and landlords to ensure a safe and legitimate rental environment.

Through the dashboard, administrators can approve or reject newly posted listings, manage user accounts, and maintain data security. This feature ensures that only verified landlords can post properties, minimizing fraudulent activity and misinformation on the platform. The administrator also has access to the system settings and audit logs, ensuring that the system operates efficiently and transparently.

For the Tenant

The Tenant Module provides renters with an accessible and reliable method for searching for verified apartments. Upon logging in, tenants can browse property listings filtered by price range, location, or apartment type. Each listing includes complete information, such as rent rates, photos, amenities, and mapped locations using Google Maps integration, allowing tenants to make informed decisions.

In addition, the system includes a real-time chat feature that enables tenants to directly communicate with landlords to inquire about availability or clarify details. Tenants can also send rental requests or reservations through the platform and receive instant notifications regarding their application status. The system further enhances convenience by maintaining a personal dashboard where tenants can track their rental history and view all past communications and approved transactions.

For the Landlord

The Landlord Module focuses on simplifying property management and tenant coordination. Landlords can register their properties in the system by adding details such as the address, rental price, and availability status. Each listing is reviewed by the administrator before being made public to ensure authenticity.

Once the listings are active, landlords can manage their tenants, track occupied and vacant units, and receive notifications of rental requests. They can communicate directly with tenants using an integrated real-time messaging tool. Furthermore, the module assists landlords in organizing tenant records, monitoring rent schedules, and receiving reminders for upcoming transactions.

This system reduces manual record-keeping and helps landlords maintain accurate and centralized information for all their properties within a single digital platform.

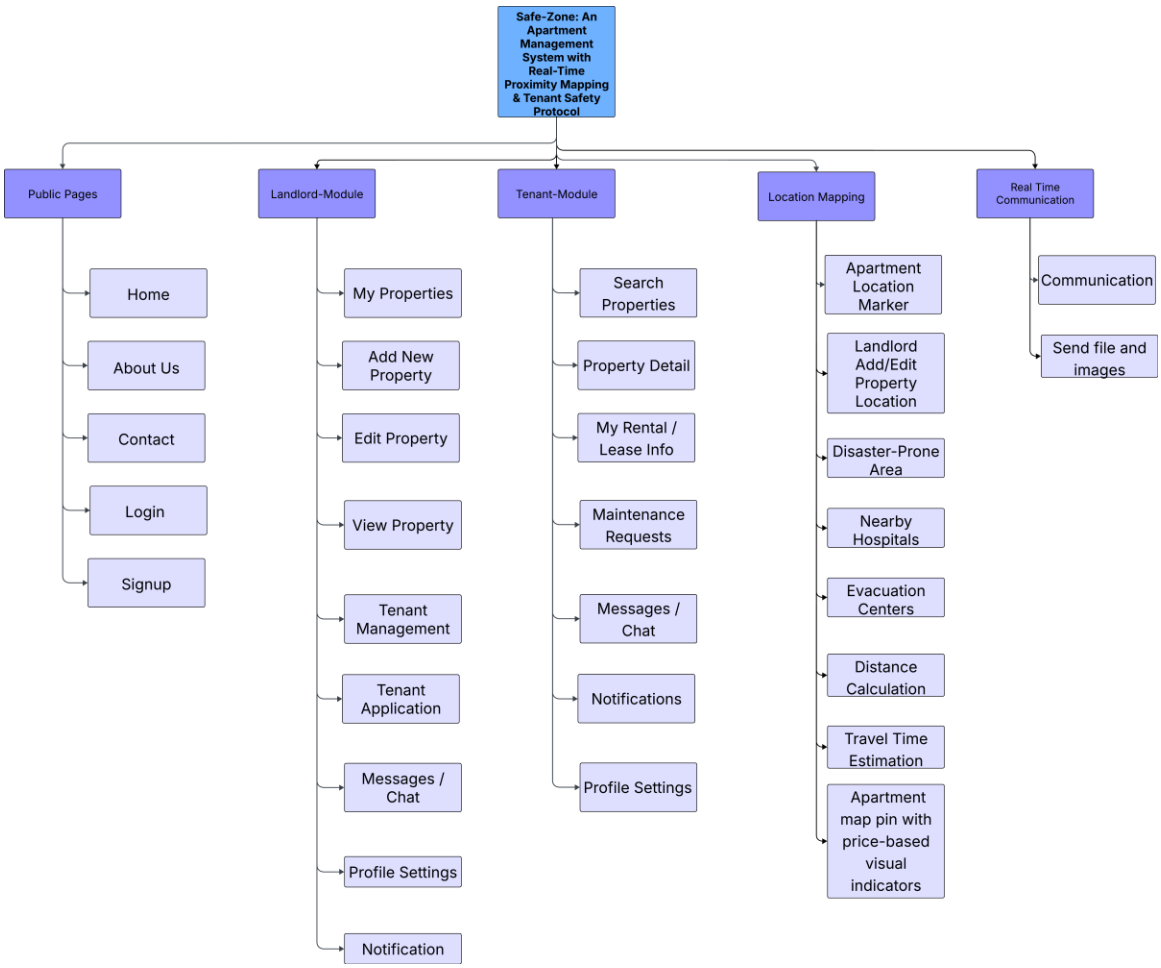


Figure 13. Project Sitemap of Safe-Zone: Apartment Management System with Real-time proximity mapping and Tenant Safety Protocol

REQUIREMENTS DOCUMENTATION

The project is a web-based apartment management platform developed specifically for landlords and tenants in San Pedro, Laguna. Its primary goal is to address the inefficiencies, unreliability, and time-consuming nature of the traditional and largely unregulated rental methods commonly used in the area.

The system provides a centralized and regularly updated database of apartment listings, ensuring that tenants have access to accurate and reliable information. It will also support real-time communication between landlords and tenants to improve coordination, reduce misunderstandings, and speed up the rental process. In addition, integrated location mapping features will allow users to easily identify apartments near preferred or accessible locations, including safer areas of the city.

By improving the overall rental search experience for tenants and offering landlords effective tools for property and tenant management, the system aims to promote greater efficiency, transparency, and trust in the local rental housing sector.

LANDLORD MODULES:

Featured: **PROPERTY PAGE**

Description: The landlord can manage all of their listed rental properties under the property page. It offers the necessary capabilities for tracking properties, updating current property details, adding new listings, and retrieving rental and tenant data.

Requirements:

Before the landlord is allowed to list any properties, the system must make sure that their identification is verified.

- The landlord must be able to create and post new property listings through the system.
- The landlord must be able to update and edit the details of their current

property listings using the system.

- The landlord must be able to review data about rental agreements and the tenants renting in their property through the system.

Use Cases:

- **Verify Landlord**: The landlord must submit valid identifying documents (IDs) for verification to activate their ability to list a property.
- **List of Properties**: The landlord can view a complete and organized list of all their currently listed or managed properties, enabling them to track the status of each property.
- **Add Properties**: The landlord can create and submit all necessary details to list a new property on the platform.
- **Edit Properties**: The landlord can modify, correct, or update the information for any of their existing properties after they have been listed.
- **Rental Information**: The landlord can access and view details pertaining to the rental status of a property, including the information about the tenant currently occupying it.

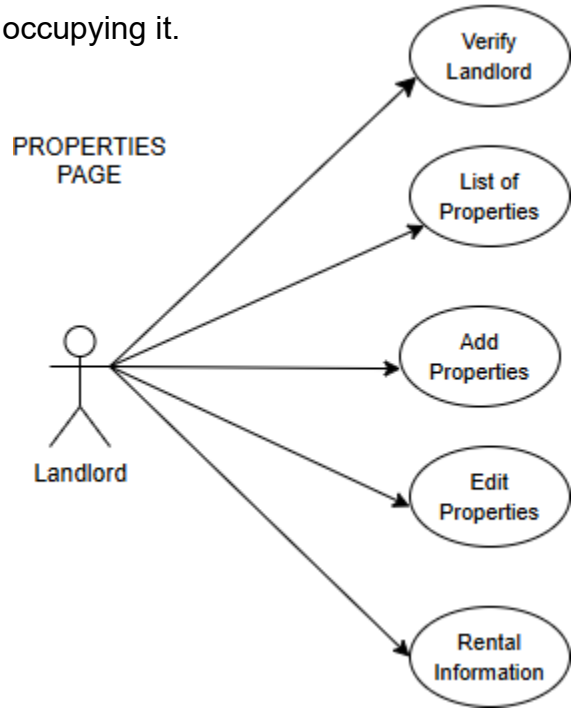


Figure 14. Use Case Diagram for Properties Page

Featured: **RENTALS PAGE**

Description: The landlord can view all their active tenant and manage their maintenance requests as well as their payment records.

Requirements:

- The system must display all the active tenants that's renting to specific landlords.
- The system must enable the landlord to manage and track the payment record of their tenants.
- The system must allow the landlord to view and manage the tenant maintenance requests.

Use Cases:

- **View Active Tenant:** The landlord can view all their tenants currently renting to their property.
- **Manage Payment Record:** The landlord can view, update and track tenant payment status.
- **Manage Maintenance Request:** The landlord can view, respond and update tenant maintenance requests.

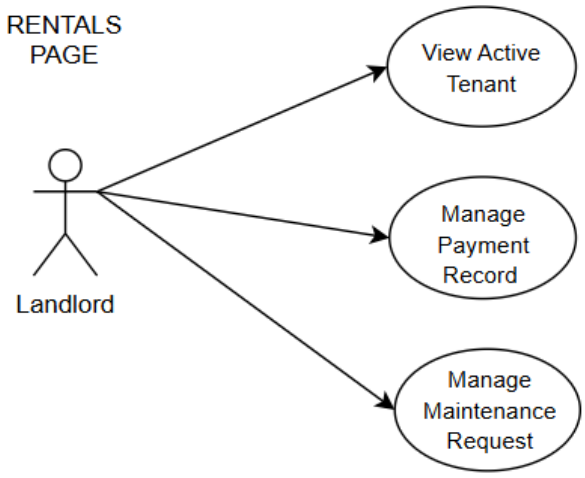


Figure 15. Use Case Diagram for Rentals Page

Featured: **MAP PAGE**

- **Description:** The Landlord can interact with their listed properties through a visual and geographic interface on the map page. The system uses Google Map API to analyze, visualize, and manage the listed apartments. Its main function is to display property information on a map so that the landlord can easily locate properties and retrieve important information.

Requirements:

- The system must visually display the geographic location of all of the landlord's listed properties on an interactive map.
- The system must enable the landlord to access the full details of the associated property by selecting a location pin on the map.
- The system must enable the landlord to see the disaster risk area.
- The system must display the color-code pin to distinguish the price range of each apartment.
- The system must display all the emergency facilities such as evacuation and hospital inside the San Pedro, Laguna.

Use Cases:

- **Location of Landlord Properties:** The landlord can view a map interface where the exact geographic location of all their listed properties is displayed using pins.
- **Disaster-Prone Area:** The landlord can see disaster prone area in the map in relation to pin their apartment location.
- **View Emergency Facility:** Landlord can view all nearby emergency facilities located near their listed apartments.
- **View Colo-Code Price Pin:** The landlord can identify the price range of each apartment base on color-code apartment pin.

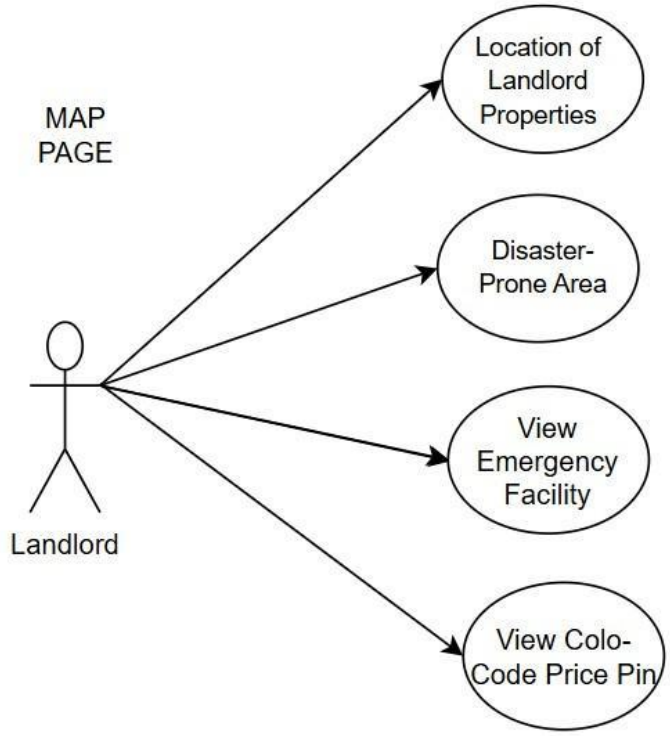


Figure 16 . Use Case Diagram for Map Page

Featured: **MESSAGE PAGE**

Description: The message page is the platform's communication hub, enabling the landlord to directly interact with their tenant. It offers real time, live interaction, and makes it easier to have specific conversations as well as general status updates, especially when it comes to important information like payments.

Requirements:

- The system must provide a channel for the Landlord to initiate and maintain communication with the tenant for status updates and general inquiries.
- The system must allow the landlord to specifically discuss or communicate information related to property payments.
- The system must support instant, real-time communication between the landlord and the tenant.

Use Cases:

- **Connect with Tenant**: The Landlord can start a message thread to communicate with the tenant, primarily for providing updates, checking status, or addressing property related inquiries.
- **Payment Information**: The landlord can use the messaging feature to send or inquire about rental payment information with the tenant.
- **Real Time Communication**: The system enables the landlord to communicate instantly with the tenant, allowing for immediate, real-time conversation like a live chat application

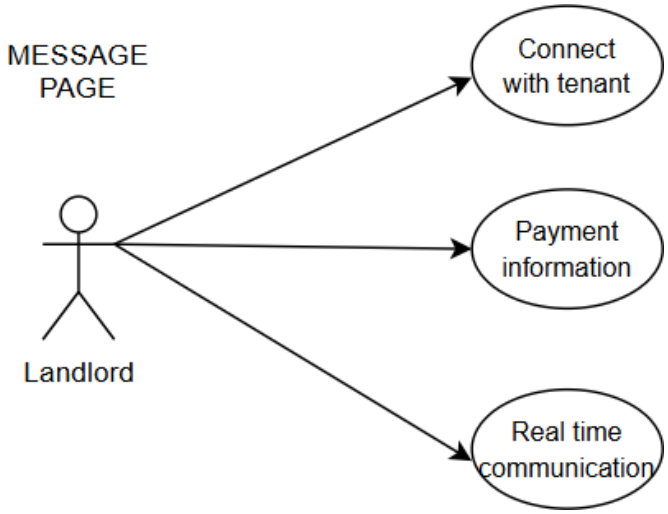


Figure 17. Use Case Diagram for Message Page

Featured: **SUPPORT PAGE**

Description: The Landlord can report technical issues, file complaints, ask for general support, and find out the company's physical location using the support page, which acts as their primary point of contact for outside assistance.

Requirements:

- The system must provide a channel for the landlord to report technical issues or bugs to the system administrators or support staff.
- The system must have a way for the landlord to submit complaints about problems with their rental properties or tenants.
- The system must make it simple for the landlord to ask for general help or more details about how to use the system.
- The system must display the company's physical address or location information.

Use Cases:

- **Contact Support**: The landlord can contact the system administrators or technical support team, primarily to report bugs or technical malfunctions within the system.
- **Complaints**: The landlord can formally submit a complaint specific problem they are experiencing with a specific tenant or an issue related to their properties.
- **Ask for Help**: The landlord can submit a request for general assistance or guidance on navigating the platform that isn't considered a bug report or formal complaint.
- **Company Location**: The landlord can view the physical address, map location, and potentially other contact details for the supporting company or management office.

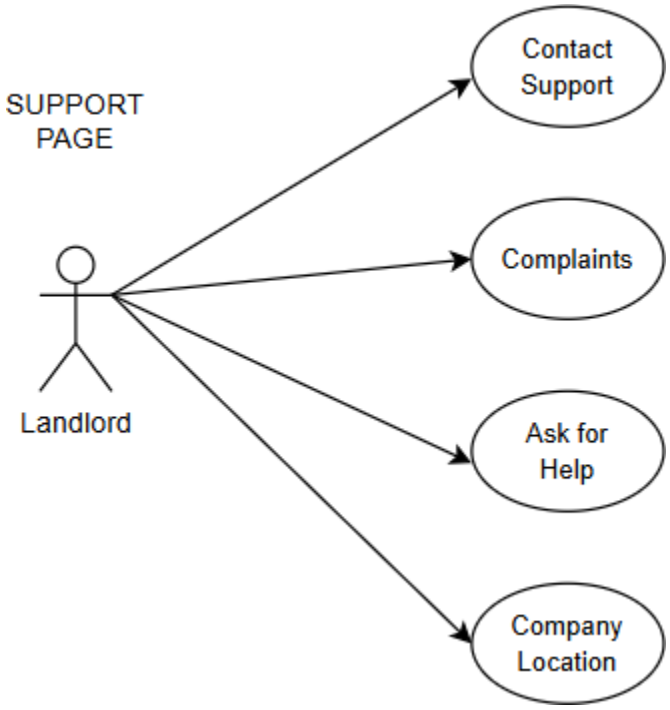


Figure 18. Use Case Diagram for Support Page

Featured: **ACCOUNT PAGE**

Description: The account page is the central location for the landlord to perform required self-service management of their user profile within the system. It grants the landlord the necessary control to maintain, update, view, and, if required, terminate their account.

Requirements:

- The landlord must be able to manage the important features of their user account with the help of the system.
- The system must allow the landlord to view all their personal and registered account details.
- The system must allow the landlord to modify and update their existing account information.
- The system must include the functionality for the landlord to permanently delete or deactivate their account.

Use Cases:

- **Manage Account**: The ability of the landlord to handle out all administrative tasks related to their user account, including the other use cases mentioned below.
- **View Ratings**: Landlords can view all the ratings and reviews made by their tenants.
- **View Active Tenants and Payments**: Landlords can manage the tenant payment by manually recording it through Active Tenants and Payment Table.
- **Landlord Information**: The landlord can access and view all their stored account details, personal data, and registration information within the platform.

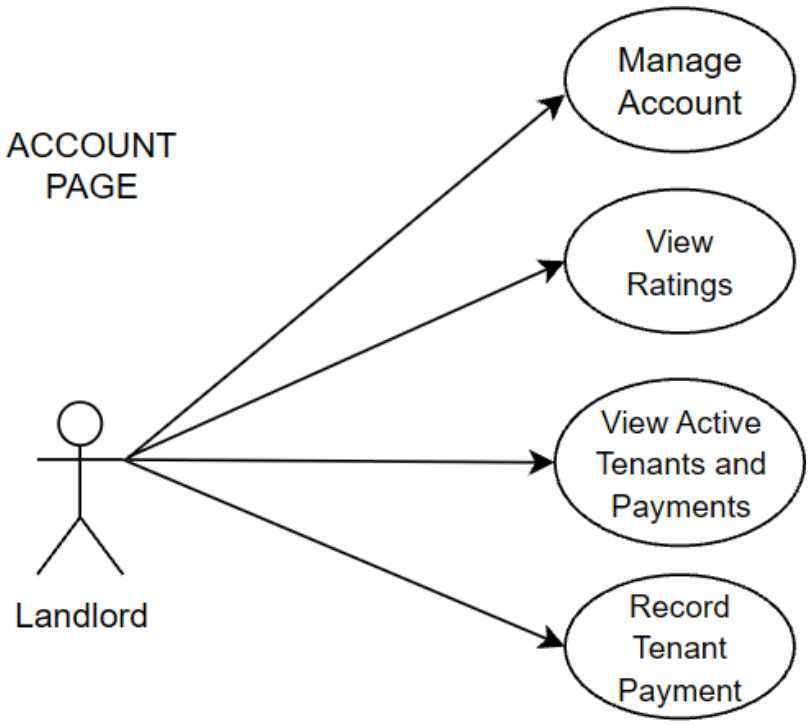


Figure 19. Use Case Diagram for Account Page

Featured: **PROPERTY DETAILS PAGE**

Description: The Landlord can view all the application requests from tenants to their specific apartment list and manage tenant application requests. In this page, the landlord can make an action such as reject the application or make a lease of agreement.

Requirements:

- The system must allow the landlord to view the details of their listed apartment.
- The system must display all tenant application requests for the selected apartment.
- The system must allow the landlord to accept the application request by creating and sending the lease of agreement.
- The system must show the status of each request based on the action taken by the landlord.
- The system must display the status of the lease agreement
- Based on whether the tenant accepts or rejects the lease of agreement made by the landlord.
- The system must remove the tenant application request if the landlord chooses to reject the application.

Use Cases:

- **View Apartment Details:** The landlord can view all the information of their listed apartment including the apartment name, price, description and address.
- **View All Tenant Requests:** The landlord can view all the tenant application requests sent to the specific apartment.
- **Create A Lease Agreement:** Landlord can choose to accept the tenant's application request by creating a lease of agreement.
    - **Reject Rental Request:** Landlords can choose to reject the

application request by the tenant.

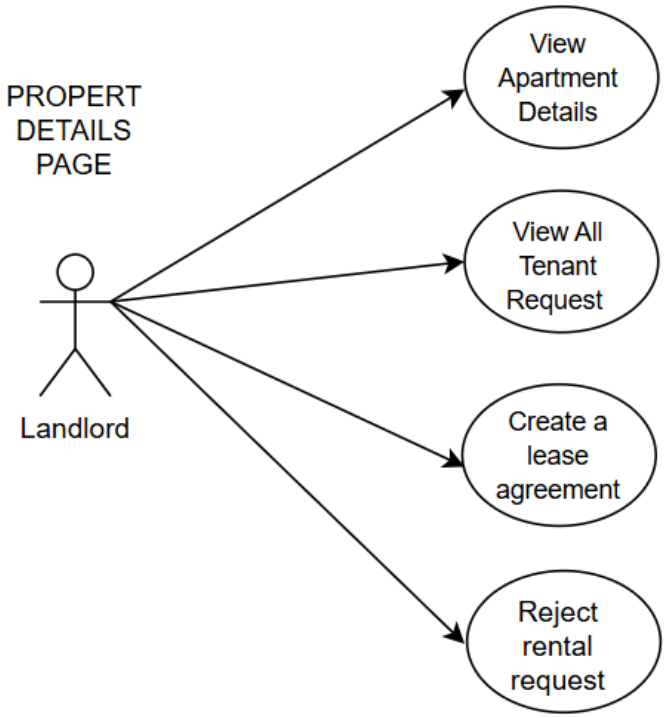


Figure 20. Use Case Diagram for Property Details Page

TENANT MODULES

Featured: **HOME PAGE**

Description: The home page is the tenant's main landing page, giving them access to the key functions for looking for, viewing, and applying to rental properties. It serves as the tenant's connection to the whole system of property listings.

Requirements:

- A complete list or feed of every property that has been posted by different landlords must be shown by the system.
- The tenant must be able to choose a property and view all related information, including pictures and complete details, through the system.
- The system must provide a clear process for the tenant to submit an official application for a chosen property.
- The system must allow the tenant to view important information about the landlord associated with a specific property listing.

Use Cases:

- **List of Properties**: The tenant can browse and view a list or catalog of all currently available rental properties posted by different landlords on the system.
- **Check Property**: The tenant can select a property from the list to view its full details, including a description, rental rate, features, location, and multiple property images.
- **Apply for Rent**: The tenant can formally start and submit a rental application for a property they are interested in, providing their personal information as required.
- **Check Landlord**: The tenant can view publicly available or relevant information about the landlord who listed a particular property, helping

they make informed decisions before applying.

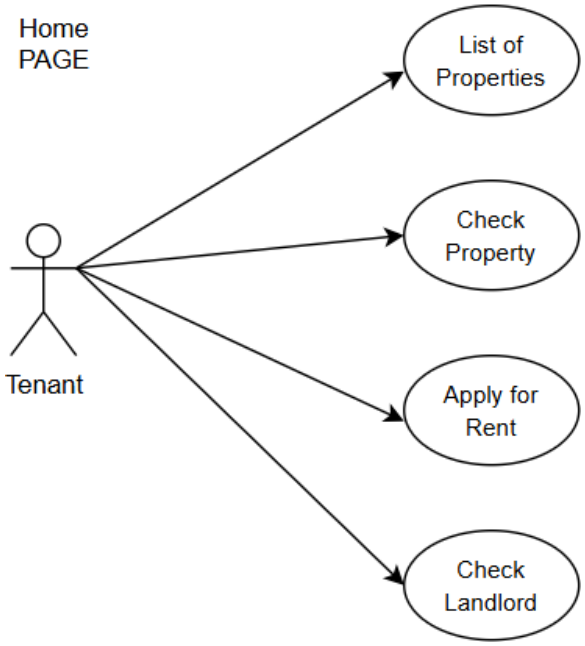


Figure 21. Use Case Diagram for Home Page

Featured: **MY RENTAL PAGE**

Description: My rental page is a private dashboard for the tenant that holds all information relevant to their current, active rental agreement. It allows the tenant to view lease agreement details, create payment records and view history, create a maintenance request, and access apartment information.

Requirements:

- The system must allow tenant to view the lease agreement information such as the terms and agreement, the start and end of contract, and the financial information related to their current rental agreement
- The system must allow the tenant to either reject or accept the lease of agreement created by the landlord.
- The system must allow the tenant to create a maintenance request.
- The system must allow the tenant to view the records and updated payment information.

Use Cases:

View Lease Agreement: The tenant has access view all the information inside the lease agreement including the lease terms; financial information related to their current rental agreement, monthly rent amount, and payment history.

- **Accept or Reject Lease Agreement:** The tenant can accept and reject the lease of agreement sent and created by the landlord.

- **Payment Record or Status**: The tenant can access and view their payment records.
- **Send Maintenance Requests**: The tenant can create and send a maintenance request to the landlord.

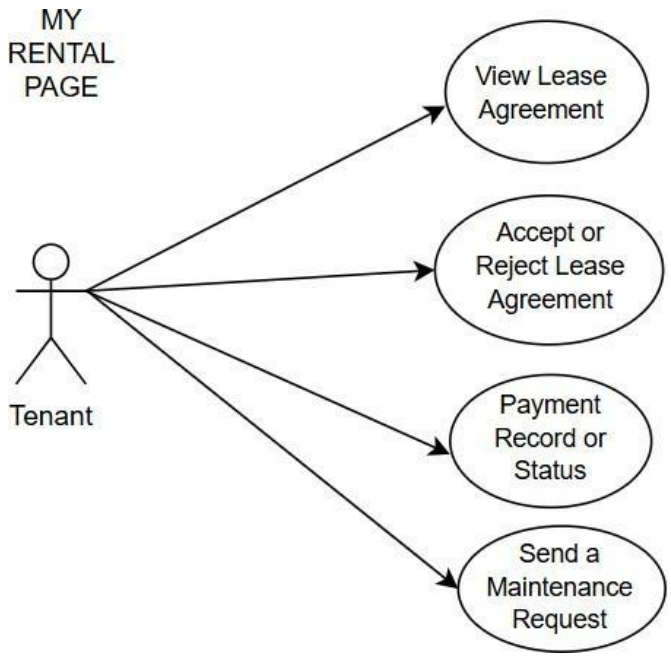


Figure 22. Use Case Diagram for My Rental Page

Featured: **MAP PAGE**

Description: The map page provides a geographical interface for the tenant to search and view available rental properties based on location. This interface allows tenants to visually see the disaster-prone area, check out nearby emergency facilities, view the color-coded price pin that indicates rental price range, and quickly access details for properties pinned on the map.

Requirements:

- The system must visually display all currently listed properties on an interactive map for the tenant.
- The system must allow the tenant to select a property marker on the map to view its full listing details.
- The system must enable the tenant to see the disaster risk area.
- The system must display the color-code pin to distinguish the price range of each apartment.
- The system must display all the emergency facilities such as evacuation and hospital inside the San Pedro, Laguna.
- The system must display the calculation of distance and estimated travel time.

Use Cases:

- **Location of Landlord Properties**: The tenant can view all available listed properties on a map to visually assess their location, accessible to other points of interest, and neighborhood context.
- **Details of Landlord Properties**: The tenant can interact with a property marker on the map to access and view full listing information, including images, rent price, and features, for that specific rental unit.

- **Disaster Prone Area:** The landlord can see disaster prone area in the map in relation to pin their apartment location.
- **View Distance and Travel Time:** The tenant can view the distance and estimated time between the selected apartment and nearby emergency facilities or landmarks.
- **View Emergency Facilities:** Tenant can view the nearby emergency facilities such as healthcare, evacuation center, fire emergencies, and earthquake rescue.
- **View Colo-Code Price Pin**: The landlord can identify the price range of each apartment base on color-code apartment pin

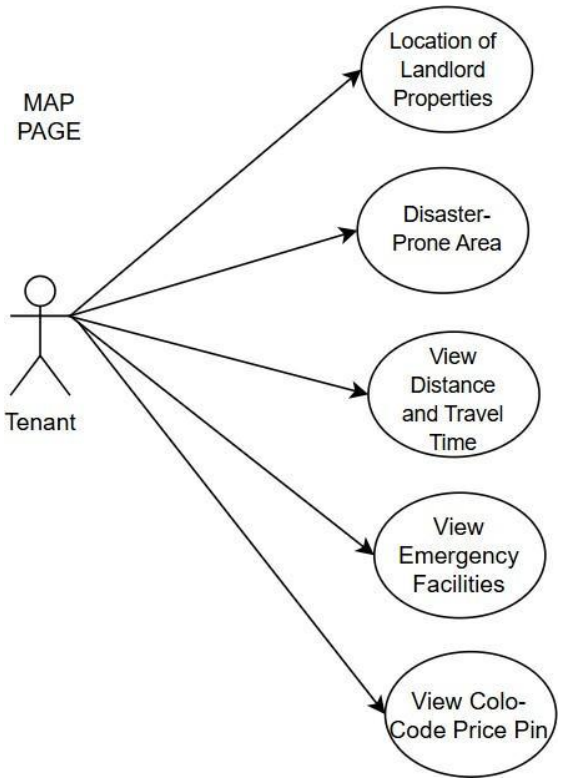


Figure 23. Use Case Diagram for Map Page

Featured: **MESSAGE PAGE**

Description: The message page provides the tenant with a direct, private communication to interact with potential or current landlords. This is essential for inquiries about listed properties and establishing contact for rental arrangements.

Requirements:

- The system must allow the tenant to start and maintain a communication link with a landlord in real time.
- The system must allow the tenant to specifically request further details or information about a property listed by the landlord.

Use Cases:

- **Connect with Landlord**: The tenants and landlords can communicate instantly and in real time about properties, applications, and other rental related topics.
- **Ask for Properties Information**: The tenant can use the messaging feature to request additional details or specific information about a property listing from the landlord.

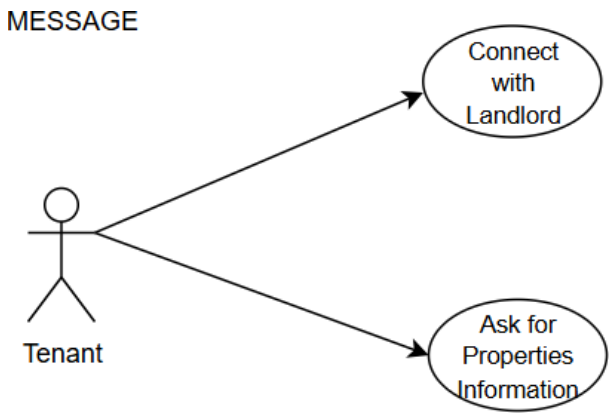


Figure 24. Use Case Diagram for Message Page

Featured: **ACCOUNT PAGE**

Description: The account page is the main location where a tenant can manage their personal and account-related settings. It provides the necessary functionalities for the tenant to maintain, update, view, and delete their account within the system.

Requirements:

- The system must allow a tenant to manage the core aspects of their account.
- The system must allow a tenant to view their detailed account and personal information.
- The system must allow a tenant to modify and update their existing account information.
- The system must provide the functionality for a tenant to permanently delete their account.

.

Use Cases:

- **Manage Account**: This is the main use case that illustrates the tenant's overall capacity to handle all account related responsibilities.
- **Edit Account**: The tenant can edit and update their existing personal and specific account information.
- **Delete Account**: The tenant can permanently delete and close their account from the system.
- **Tenant Information**: The tenant can view all of their stored information and personal data in the system.

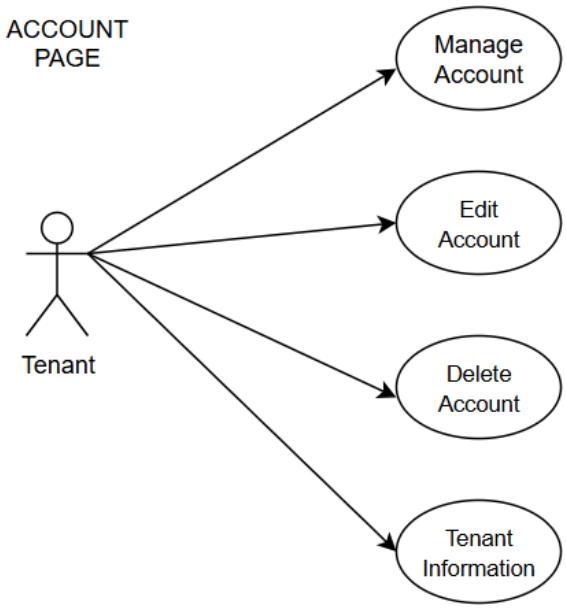


Figure 25. Use Case Diagram for Account Page

Featured: **SUPPORT PAGE**

Description: The Tenant's primary resource for requesting help, reporting problems, submitting complaints, and locating company information is the support page. It ensures that the tenant gets multiple options for resolving issues and receiving assistance within the system.

Requirements:

- The system must give the tenant a way to communicate with support personnel, like email, to report bugs or technical difficulties.
- The system must give the tenant a way to officially submit reports or complaints about the actions or account of a landlord.
- The system must make it simple for tenants to ask administrators for general help when they're experiencing issues with the system.
- The physical address or location details of the business must be shown by the system.

Use Cases:

- **Contact Support**: The tenants can report system bugs or other technical issues they come across by contacting the technical support team, usually through email.
- **Complains**: The tenant has the choice to formally file a report or complaint about the conduct or behavior of a specific landlord's account.
- **Ask for Help**: The tenant can submit a request for general assistance or guidance on navigating the platform that isn't considered a bug report or formal complaint.
- **Company Location**: The tenant can view the physical address, map location, and potentially other contact details for the supporting company or management office.

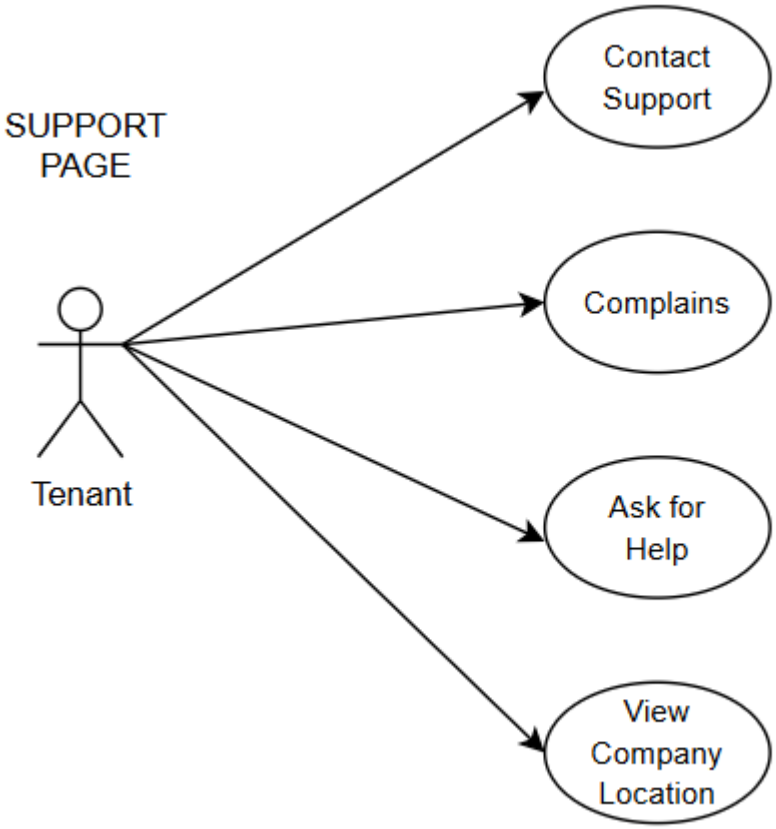


Figure 26. Use Case Diagram for Support Page

ADMINISTRATOR MODULES:

Use Case Diagram: Account Management Module

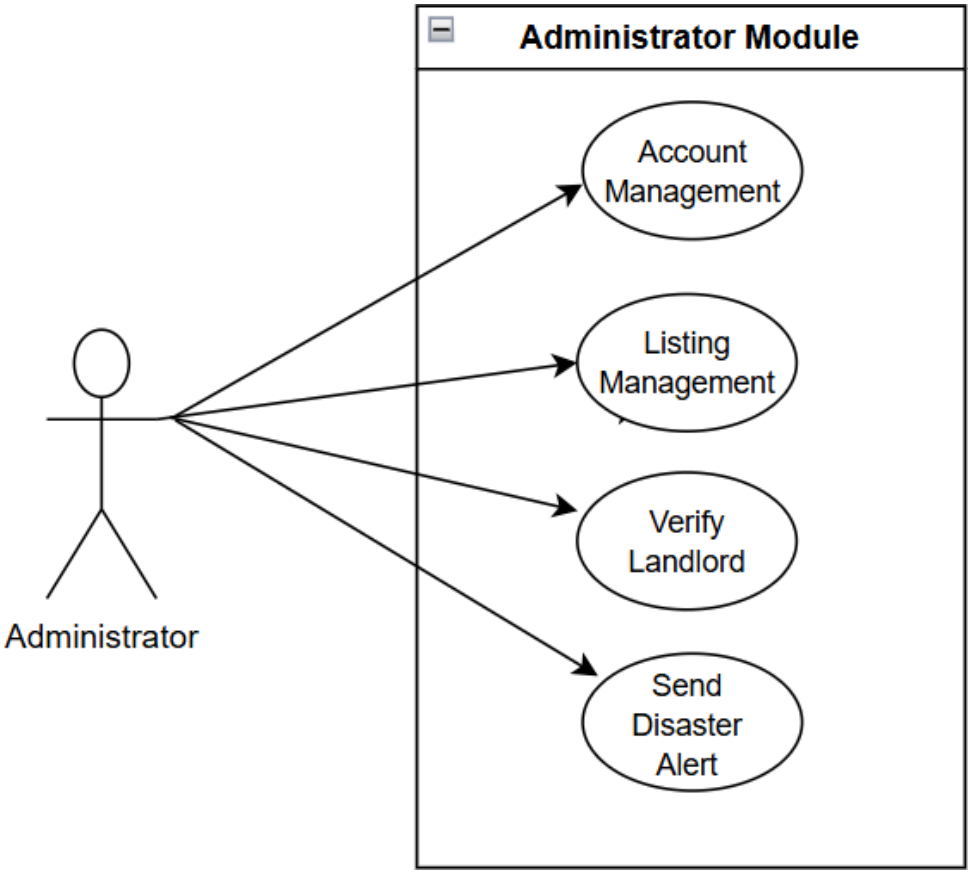


Figure 27. Use Case Diagram for Administrator Module

Use Case 1: Account Management

Actors:

- Administrator

Description:

The Administrator can manage all landlord and tenant accounts within the system. This includes the ability to create, view, modify, activate, deactivate, or delete user accounts.

Preconditions:

- The Administrator must successfully log in to the Administrator Module after authenticating.
- The Administrator's user profile must own management privileges for both landlord and tenant account data.
- The system must contain active landlord and tenant accounts that are available for management.

Flow:

1. The Administrator navigates to the Account Management section.
2. The system displays a list of all user accounts (Landlords and Tenants).
3. The Administrator searches for or selects a specific user account.
4. The system displays detailed account information.
5. The Administrator selects an action (Edit, Activate, Deactivate, View Activity Log).
6. The system executes the chosen action and confirms the result to the Administrator.

Alternative Flow:

1a. If the account searched for does not exist, the system informs the Administrator and returns to the account list.

1b. If the Administrator attempts to deactivate or delete an account, the system requests confirmation before proceeding.

Use Case 2: Listing Management Actors:

- Administrator

Description:

The Administrator can oversee and control all property listings in the system. This includes the ability to view, approve, edit, or remove listings to ensure they comply with platform policies and maintain system honesty.

Preconditions:

- The Administrator must successfully log in to the Administrator Module after authenticating.
- The Administrator has the necessary privileges to access and modify property listing information.

Flow:

1. The Administrator navigates to the Listing Management section.
2. The system displays a list of all property listings, typically prioritizing those pending reviews or those reported by users.
3. The Administrator searches for or selects a property listing for review.
4. The system displays all property details, images, and the status of the listing.
5. The Administrator reviews the listing against platform policies.

6. The Administrator selects an action (Approve, Reject, Delete).
7. The system updates the listing status or information and sends a notification to the associated Landlord.

Alternative Flow:

2a. If the Administrator selects a listing that has been reported, the system also displays the details of the user's reports for review.

2b. If the listing violates policy, the Administrator rejects or deletes the listing, providing a reason that is recorded in the system.

Use Case 3: Verify Landlord Actors:

- Administrator

Description:

The Administrator is responsible for reviewing and authenticating the identity of new landlords who have submitted verification documents to ensure they are legitimate users and authorized to list properties.

Preconditions:

- The Administrator must successfully log in to the Administrator Module after authenticating.
- There are one or more Landlord accounts awaiting verification.

Flow:

The Administrator navigates to the Verify Landlord section (a queue of pending verifications).

1. The system displays a list of landlords who have submitted documents for verification.
2. The Administrator selects a landlord from the list.
3. The system displays the landlord's submitted profile information and uploaded verification documents.
4. The Administrator reviews documents for authenticity and validity.
5. The Administrator selects either verify or reject verification.
6. The system updates the landlord's account status to verified or verification rejected and triggers an automatic notification to the landlord.

Alternative Flow:

3a. If the documents are invalid or unclear, the Administrator rejects the verification.

Use Case 4: Send Disaster Alert Actors:

- Administrator

Description:

The Administrator is responsible for sending disaster alert to notify the tenants and landlords about the disaster such as typhoon, flood, earthquake, and fire incident. The system will send a notification especially around the affected area, the admin will also be responsible for sending these disaster alerts through email.

Preconditions:

- The Administrator must successfully log in to the Administrator Module after authenticating.
- Disaster information such as typhoons, floods, earthquakes, and fire incidents must be declared.

Flow:

The Administrator navigates the Disaster Alert section.

1. The system will display all disaster types.
2. The administrator will select the disaster type.
3. The administrator will enter alert details such as the severity of the disaster, description, and advisory message.
4. The administrator will confirm the disaster alert.
5. The system will send in-page notifications and email alerts.
6. The system will log the disaster alert for record and monitoring purposes.

Alternative Flow:

4a. If no apartments are affected by the selected disaster area:

- The notification will not be sent by the users.

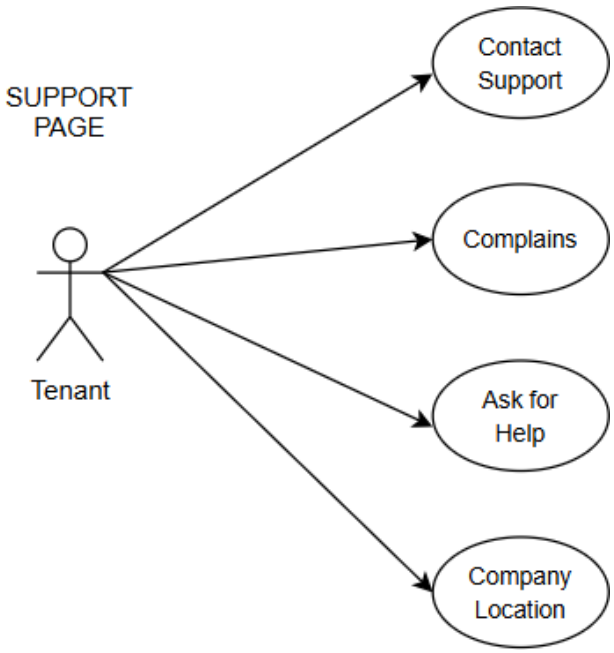


Figure 28. Use Case Diagram for Support Page

DESIGN OF SOFTWARE, SYSTEMS, PRODUCT AND/OR PROCESS

This figure aims to explain the meaning of terms and concepts used to analyze web application needs and to discover the appropriate relationships between various concepts.

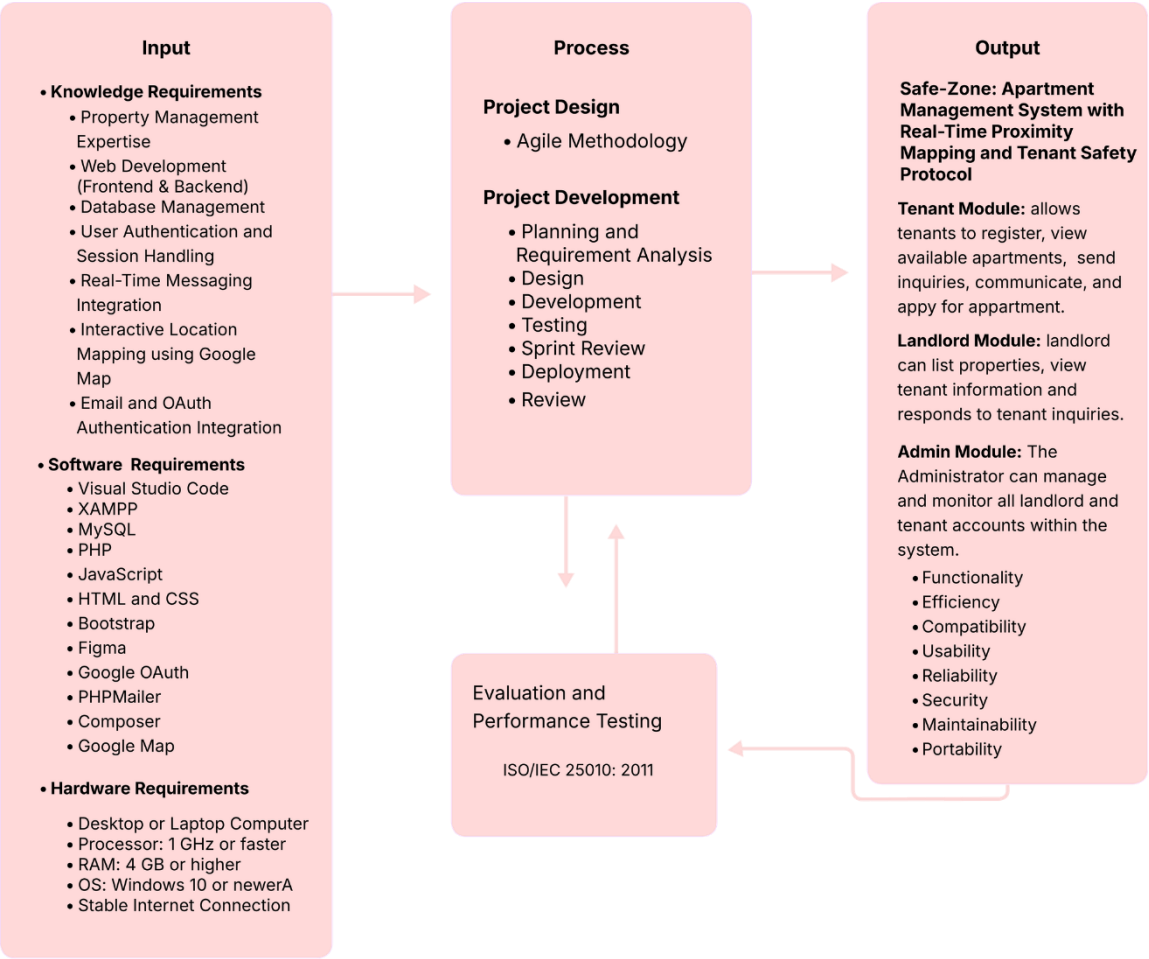


Figure 29. Conceptual Model of the Study

The figure shown above is an illustration of the Input-Output-Process (IOP) diagram, which provides a clear overview of the system. This diagram serves as a blueprint of the system, demonstrating how the web-based apartment management platform aligns with the system requirements during the development and completion of the project. It is organized into three components: Input,

Process, and Output. The first, Input, included all required knowledge needed to complete the project, software tools to compile all the necessary features of the system, and hardware requirements with the minimum system specifications and a stable Internet connection to support development and testing activities. The next step is the use of the Agile software development methodology to initiate all the requirements in the project design, with specific phases such as planning, design, development, testing, sprint, deployment, and maintenance. Lastly, the output is A Web-based Apartment Management Platform Incorporating Tenant and Landlord Features with Real-Time Communication and Location Mapping to enhance the experience of both tenants and landlords in finding apartments in a secure way and listing their properties in a more manageable way. This system will be evaluated using the ISO/IEC 25010:2011 software quality standard.

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves within the Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol. It shows the flow of information between users such as administrators, landlords, and tenants, as well as the system processes and data storage. The DFD presents both the overall system interaction and the detailed processes involved in managing accounts, listings, and safety features. This helps provide a clear understanding of how the system operates and ensures efficient and organized data handling.

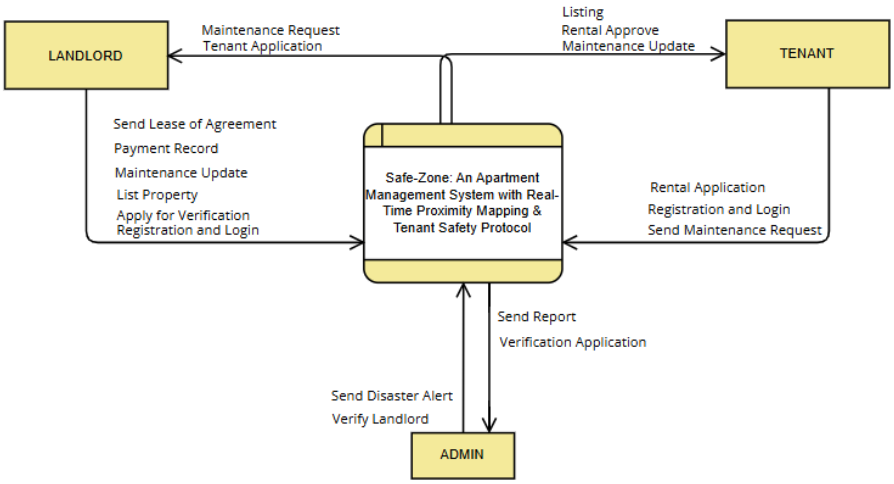


Figure 30. Data Flow Diagram 0 of Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol

This data flow diagram illustrates the triadic actor structure of Landlord, Tenant, and Admin within the "Safe-Zone" apartment management system, delineating their respective interactive nodes. The system integrates core operational workflows such as rental processing, maintenance coordination, and safety protocols, with bidirectional flows indicating transactional and

communicative exchanges between entities. Administratively, the diagram emphasizes a centralized oversight mechanism where the Admin module governs verification, alert dissemination, and report management to ensure systemic integrity and tenant safety.

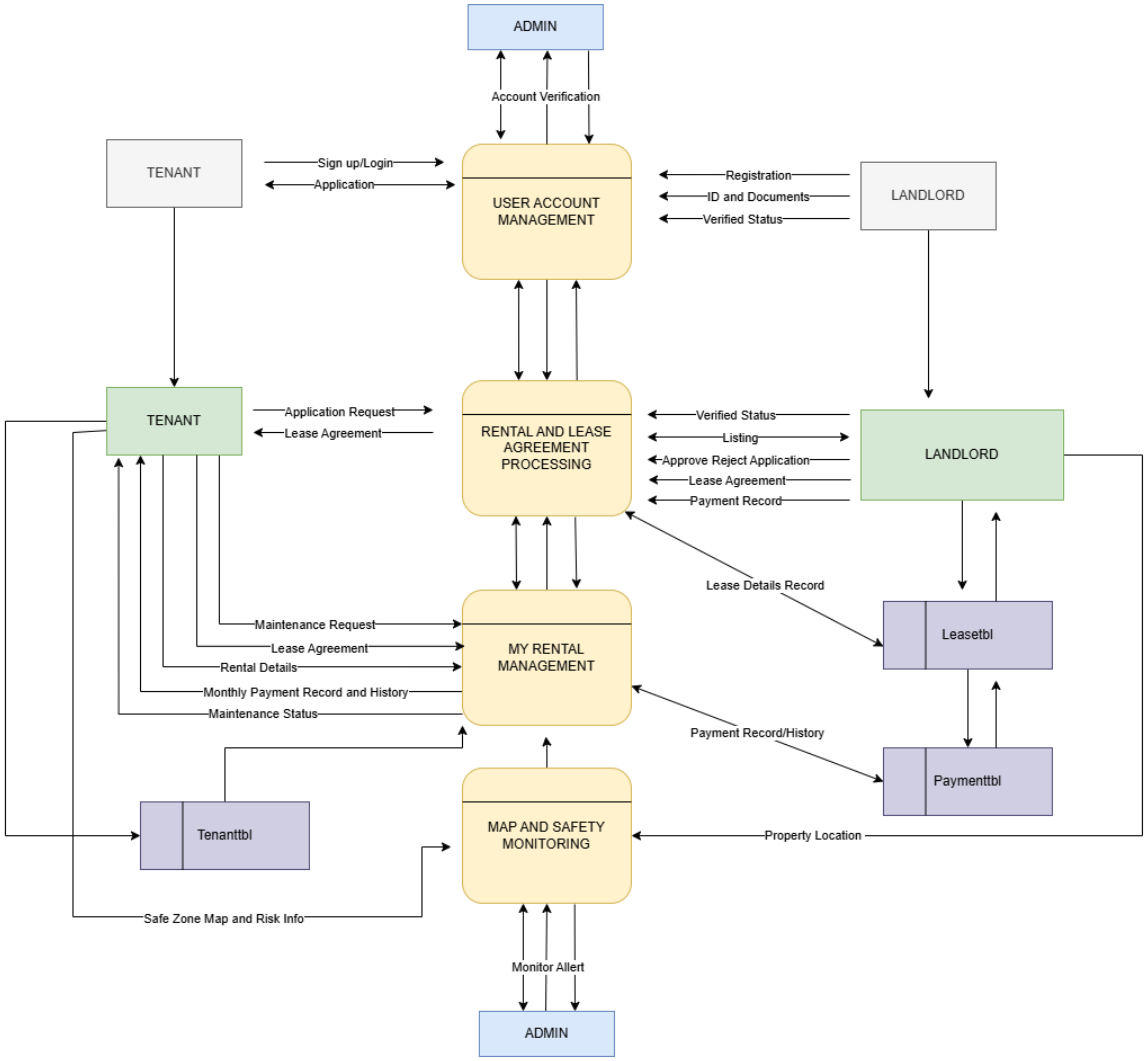


Figure 31. Data Flow Diagram Level 1 of Safe-Zone: An Apartment Management System with Real-Time Proximity Mapping & Tenant Safety Protocol

This level 1 data flow diagram decomposes the core processes of the Safe-Zone system into four primary functional components: User Account Management,

Rental and Lease Agreement Processing, MyRental Management, and Map and Safety Monitoring. The diagram illustrates how tenant applications undergo verification by the administrator, with the resulting verified status enabling the landlord to generate lease agreements and manage rental listings. It further shows the integration of payment records and maintenance requests within the MyRental Management module, linking tenants, landlords, and payment history through structured data tables such as Leasetbl, Paymenttbl, and Tenanttbl. The Map and Safety Monitoring component incorporates property location data and risk information to support real-time proximity mapping and tenant safety protocols. Overall, the diagram demonstrates a structured flow of data across entities, highlighting the interconnected roles of tenants, landlords, and administrators in managing rental operations and safety features.

Development and Testing

The agile methodology is a procedure for designing, delivering, and testing high-quality software at the lowest cost and in the shortest possible time. The heart of the process centers around iterative development, namely allowing teams to deliver faster with a greater ability to react to change.

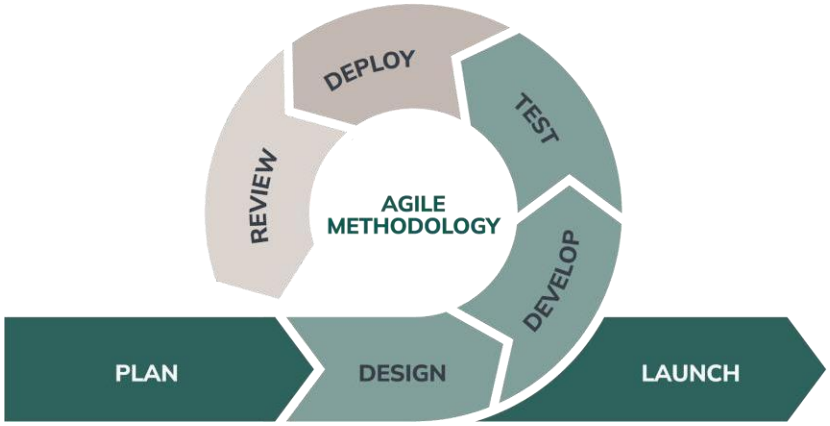


Figure 32. Agile Methodology

- Planning and Requirement Gathering: This phase involves identifying and defining the functional and non-functional requirements of the system. The team collaborates with stakeholders to gather information, and deliverables are clearly outlined. User stories are also created to define specific tasks for each sprint.

- Design:In this phase, the system architecture and interface design are planned. Wireframes and prototypes are created to visualize the flow of the system. The database structure and API integrations are defined. The design phase ensures that both the front-end and back-end components are aligned with the project's goals and user requirements.
- Development: This stage involves building the actual system in sprints. Each sprint focuses on a set of prioritized features — such as tenant registration, property listing, messaging, and location mapping. Developers work on front-end and back-

end components while ensuring proper integration with databases and external APIs. Features are developed incrementally so that functional modules can be tested immediately after completion.

- Testing: Continuous testing helps detect and fix bugs early, ensuring stable builds after each sprint. Testing is performed throughout each sprint rather than after full system completion. This includes:

    a. Unit Testing to verify individual components,
    b. Integration Testing to ensure smooth interaction between modules, and
    c. Functional Testing to validate system behavior against user stories

- Sprint Review: After every sprint, a review meeting is conducted with stakeholders. The development team demonstrates the completed features, gathers feedback, and identifies areas for improvement. Any changes or additional requirements are documented and added to the product backlog for the next sprint cycle. This step ensures that the system evolves according to actual user needs.

- Deployment: Once the system meets the required functionality and quality standards, it is deployed for actual use. Ongoing maintenance and updates are performed to ensure stability, security, and scalability.

- Review: After deployment, a final review and evaluation are conducted to assess the overall performance, usability, and quality of the system. The evaluation follows the ISO/IEC 25010:2011 software quality model, focusing on functionality, efficiency, compatibility, usability, reliability, security, maintainability, and portability. Feedback gathered during this stage is used for further improvements, system maintenance, and future development. Throughout the development process, collaboration and communication were maintained through regular meetings, feedback sessions, and testing cycles. This iterative approach, in

addition to allowing for quick adaptation and continuous quality improvement, helped ensure that the system effectively achieved the objectives it was designed for.

The Agile methodology that embedded iterative development, continuous testing, and ongoing stakeholder engagement to allow for more timely delivery of a reliable final product.

TESTING PROCEDURE

The Apartment Management Platform underwent Alpha and Beta Testing to ensure it meets design specifications and user requirements.

Alpha Testing:

Alpha testing is a type of acceptance testing performed internally by the organization developing software.

- Evaluate the platform in a controlled development environment using local servers and dedicated tenant and landlord test accounts.
- Select a small group of developers, researchers, and technical advisers to perform the initial testing and provide detailed performance observations.
- Conduct comprehensive testing of all major features, including account management, property listing, tenant–landlord chat, and location mapping.
- Assess the system's functionality, usability, performance, and stability to ensure efficient operation across all devices and browsers.
- Collect alpha tester feedback via forms, interviews, or direct communication to identify technical errors, design issues, and usability challenges.
- Review and resolve identified issues by debugging, enhancing the UI, and optimizing the responsiveness of chat and mapping components.
- Implement improvements based on feedback to prepare the platform for

external

Beta Testing:

Beta testing is the stage of software testing that involves releasing the application to a larger group of external users.

- Deploy the platform in a real-world environment, allowing actual tenants and landlords to access the system via desktop and mobile devices.
- Identify a target group of beta testers, including a limited number of property owners and tenants, to evaluate the platform's usability, reliability, and user experience.
- Allow beta testers to freely use the system for real tasks, including creating accounts, listing/browsing apartments, chatting, and viewing map locations.
- Observe and record the system's performance, responsiveness, and stability during real-time operations, ensuring smooth functioning of communication and mapping under normal usage.
- Gather beta user feedback via online forms, interviews, or in-app tools to capture opinions, suggestions, and encounter issues.
- Analyze collected feedback to identify common problems and improvement areas, such as UI adjustments, chat responsiveness, or map loading optimization.
- Apply necessary refinements and updates based on user insights, ensuring the final platform meets technical standards and user expectations before full deployment.

Research Evaluation Instrument

The researchers utilized the ISO 25010 Software and Data Quality Standard. This international standard, specifically ISO 25010:2011, provides a detailed model that defines the characteristics and sub-characteristics of software product quality.

The system will be assessed against eight key quality levels defined by the ISO 25010 standard: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

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